

Fixed-height shifted rectangles

An unconditional proof of the A181199 asymptotic,
all-order expansions, and an algebraicity dichotomy

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For every fixed number of rows m , let $T_m(n)$ count the $m \times n$ arrays increasing in rows, columns, diagonals, and downward antidiagonals. The principal result is

$$T_m(n) \sim \frac{\sqrt{m} \prod_{j=0}^{m-1} j!}{2^{m(m-1)} (2\pi)^{(m-1)/2}} \frac{m^{mn}}{n^{(m^2-1)/2}}.$$

The proof supplies every inverse-power correction, proves the asymptotic conjecture in OEIS A181199, and implies that $\sum_{n \geq 0} T_m(n) z^n$ is algebraic if and only if $m \leq 2$.

Abstract. We split the order-polytope model of a shifted rectangle at a fixed numerical threshold. Except for an explicitly bounded exponentially small boundary contribution, its enumeration is an expectation under independent binomial random variables, weighted by a squared Vandermonde and a rational pair interaction. The shifted hook formula identifies the weight exactly. Standardized binomial moments are polynomials in $1/n$, so the expectation has an effective all-order expansion with rational coefficients. Its leading Gaussian integral gives the constant above; the first two corrections are proved as explicit polynomials in m . We obtain the conjectured constant $9\sqrt{5}/(2^{17}\pi^2)$ for A181199 and five correction terms, a universal limiting ratio to ordinary rectangular tableaux, nonalgebraicity at every height at least three, and Gaussian-ensemble laws for random intermediate row lengths. Complete proofs, an exact-arithmetic coefficient engine, independent enumerators, and a source/status audit are included.

Status. AI-assisted, unrefereed research manuscript. The theorems have proofs below, but have not been independently peer reviewed or proof-assistant verified. Historical priority is not certified. No conjectured recurrence is used as a premise, and none is claimed proved here.

[write] **Added 5 October 2026, batch 98: two Parts.** Since 5 October 2026 this report has two Parts. Part I (Sections 1–13 and Appendices A–B) is the batch-85 manuscript described in the first note of Section 1.3, printed as before: its sections, equations, theorems, research questions and labels keep their numbers, and that note’s “a single manuscript, printed in full” now describes Part I. Part II (Sections 14–25 and Appendices C–D) is manuscript 02 of batch 98, *Rational diagonals and rare boundary phases in shifted rectangles* (4 October 2026, pinned at db20eb379), printed in full with the label prefix `shr:bp:`. It proves that F_m is the diagonal of a rational power series, hence D-finite, for every fixed m (Theorem 14.1), which answers the first half of Research question 7; the second half, the growth of the minimal order, stays open. It also proves a sharp all-order expansion for the number $R_m(n)$ of arrays whose first row is completed before the last row starts (Theorem 14.2). That count is a small part of the boundary error of Theorem 4.1, not its asymptotic, so Research question 4 is not answered by it (Remark 14.4); the leading term of the boundary error is the write’s Proposition 18.3. Dated notes at the end of Section 4.1, after Remark 8.1 and at Research questions 4 and 7 say so.

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Part I

An unconditional proof of the A181199 asymptotic, all-order expansions, and an algebraicity dichotomy

1 The problem, the open target, and the source audit

For positive integers m, n , consider arrays $(a_{i,j})$ containing each integer in $\{1, \dots, mn\}$ exactly once, subject to

$$a_{i,j} < a_{i,j+1}, \quad a_{i,j} < a_{i+1,j}, \quad (1.1)$$

$$a_{i,j} < a_{i+1,j+1}, \quad a_{i,j+1} < a_{i+1,j}, \quad (1.2)$$

whenever the indicated cells exist. Write $T_m(n)$ for their number, and set $T_m(0) = 1$. The diagonal inequality is redundant once row and column increase are imposed. The downward-antidiagonal inequality is not.

These numbers form the array A181196. Its fixed-height rows include A181197 for $m = 3$, A181198 for $m = 4$, and A181199 for $m = 5$ [1, 2, 3, 4]. Throughout this article, m is fixed and the row length n tends to infinity. Interchanging these two parameters changes the problem: already $T_2(n) = C_{n-1}$ for $n \geq 1$, whereas $T_m(2) = 1$ for every m .

1.1 The conjecture resolved here

The A181199 entry inspected on October 3, 2026, labels the following formula as a conjecture of Vaclav Kotesovec, dated February 27, 2023:

$$T_5(n) \sim \frac{9 \cdot 5^{n+1/2}}{2^{17} \pi^2 n^{12}}. \quad (1.3)$$

The entry says it was based on Christoph Koutschan's conjectured recurrence. Kauers and Koutschan discuss the relevant guessed recurrences and finite-sum conjectures for heights four and five in their 2023 article [7, 4]. We do not assume those recurrences. We prove (1.3) directly from the combinatorial definition and extend it to every fixed height.

1.2 What is established background, and what is contributed

The shifted hook product and the order-statistics interpretation are classical tools, not new inventions of this report. Sun's treatment of shifted strips gives both tools and proves enumeration results in several small cases [9]. The known height-three leading asymptotic in A181197 is recovered, not newly conjectured. Chan's transfer-matrix theory concerns repeating strips with a fixed number of cells in each row; in the notation here that is the regime with n fixed and m growing [10]. Its constant-coefficient recurrences do not supply a fixed-height asymptotic in n .

The contributions developed in this manuscript are the exact random-threshold reduction with a boundary bound, the resulting all-height all-order theorem, universal first and second corrections, an algebraicity classification, and intermediate-shape limit laws. These statements are proved below. The source review did not locate these statements in the works inspected, but that observation is not an exhaustive priority search. In particular, a formula's continued "conjecture" label in OEIS is not by itself proof that no published proof exists.

1.3 Relationship to ProveIt

The repository was inspected at revision `6bf7f30d0352f7596e70928b3d4f304914075907`. Identifier searches for A181198 and A181199 returned no matching repository report. This is a limited search result, not a claim to have audited every repository file. The existing path-forest reports distinguish unconditional expansions, conjectured recurrences, and inverse-index approximations [15]. We retain that separation. No theorem in that report is a mathematical dependency of the present proof, and the standard inverse-index method in Section 10 is not presented as new.

Scope of the resolution. Equation (1.3) is proved here. The order-two recurrence for A181198 and the order-three recurrence for A181199 are *not* proved here. The asymptotic series is an all-order Poincaré expansion, with a fixed truncation order; no Borel summability, Stokes data, or complete exponentially improved transseries is asserted.

[write] Added 3 October 2026, batch 85. This report is a single manuscript, printed in full: manuscript 09 of batch 85 of ProveIt’s incoming-reports intake, delivered as the archive `Shifted_Rectangle_Asymptotics.zip` (343,338 bytes, wrapper directory `Shifted_Rectangle_Asymptotics/`) in commit `9d6968c8a`, and placed in commit `ddf8df5d5` in the research-report collection, among the OEIS-sequence asymptotics reports, as the report `a181199-shifted-rectangles`. The title page reads “Prepared for Vladimir Reshetnikov”; the PDF metadata names “Research report prepared for Vladimir Reshetnikov”. No human author and no tool are named. The delivered text is printed unchanged except that every label carries the prefix `shr:`, the title page no longer creates a duplicate page anchor, and dated notes such as this one are added.

Pin and repository claims. The revision quoted above, `6bf7f30d0` (3 October 2026), is the ProveIt commit the manuscript was written against. At placement, and again at this writing, the intake searched the whole tracked tree for A181196–A181199, shifted rectangles, shifted tableaux, shifted hook products and Sun’s article; apart from the arrival archive, the only hits are the batch-85 sibling named below, which points to this report. This report continues no repository material, and the sentence “Identifier searches for A181198 and A181199 returned no matching repository report” remains true. “The existing path-forest reports” is one report, `oeis-sequence-asymptotics/a189281-path-forest-expansions`, in three Parts; as the manuscript says, none of its theorems is used here. No statement of this report is formalized in Lean or Rocq, no formal development in ProveIt treats shifted tableaux, and the report’s place in the collection confers no formal status.

Neighbours. The companion reports of the same batch are `a047874-long-increasing-subsequences` (manuscript 08), whose fixed-sector problem also uses the hook count of the rectangle with m rows of length n as a normalizer, and `a182220-source-boundary` (manuscript 07, unrelated mathematics). Other conjectures of the 2023 article of Kauers and Koutschan [7] are proved in the collection reports `enumerative-combinatorics/a181280-binary-matrix-formula` (its Conjecture 20, Section 6.5, a C-finite formula), `enumerative-combinatorics/a195806-hexagonal-lattice` (Conjecture 11) and Part IV of `a215561-fixed-composition-excursions` (Conjecture 15). The present report concerns that article’s Section 6.4, and leaves the guessed recurrences of that section unproved. Two notes below relate Corollary 2.4 and Section 10 to the transseries volumes in `Analysis/Transseries/docs/series-and-transseries/`.

Notation. The manuscript reuses several letters with local meanings. They are kept as delivered; the table fixes each reading. No symbol was renamed.

[write] Notation table of the note in Section 1.3 (added 3 October 2026)

Symbol	Meanings in this report
$T_m(n)$	the shifted-rectangle count (A181196 family), <i>not</i> the number $f^{(n^m)}$ of standard Young tableaux of the $m \times n$ rectangle; the transseries volume writes $T_d(n)$ for the latter, with d rows.
d	$\binom{m}{2}$, the number of row pairs ((2.1)); in the transseries volume d is the number of rows, m here.
$K_m; K; K_j(X)$	the leading constant (2.2); $K = K_m$ in Section 10, but the number of samples below p in the proof of Theorem 9.2; Krawtchouk polynomials in Section 6.1.
$L; L_5(n)$	a truncation order (Theorem 2.1); $\log(y/K)$ in Section 10; the leading term in Section 7.3.
$h_r(q); h_j$	complete homogeneous polynomials (5.3); squared norms $j! n^j/n^j$ in Section 6.1.
λ	a state or strict partition (Section 3, Appendix A); $m \log m$ in Section 10.
$R_n; R_m; R(z); \mathcal{R}_{L,n}$	the pair weight (4.2); the radius m^{-m} in Section 8; the rational function of Appendix A; the Taylor remainder (5.12).
$p; p_r$	the threshold in $(0, 1)$; the power sums $\sum_i Y_i^r$ (Section 6).
$q_{ij}; q$	$(y_i + y_j)^2$; a derivative order in Section 8.
$Z_m(n); Z_i$	the binomial Vandermonde normalizer (6.1); standardized counts in Section 9.
$A, B; \mathcal{A}$	$\binom{m}{3}$ and $m(m-1)^2(m-2)/12$ (Section 6.1); the event that all comparisons hold.
$C(z); C_{n-1}; C_{m,L}; C_n$	the Catalan generating function; Catalan numbers; a constant in (5.12); the central event.
$E_{m,n}(p); \mathbb{E}; \mathcal{E}_z$	the boundary error (4.3); expectation; the moment functional (5.8).
$H(\lambda); H_j; H_{20}$	the hook expression of Appendix A; Hermite polynomials; the constant (6.11).
β_j	recurrence coefficients (Section 6.2); not the exponent $\beta = -\alpha_m$ of the transseries volume.

2 Main results

Set

$$d = \binom{m}{2}, \quad J_m = \prod_{j=0}^{m-1} j!, \quad M_m(n) = \frac{(mn)!}{(n!)^m}, \quad \alpha_m = \frac{m^2 - 1}{2}, \quad (2.1)$$

and define

$$K_m = \frac{\sqrt{m} J_m}{4^d (2\pi)^{(m-1)/2}}. \quad (2.2)$$

The multinomial coefficient $M_m(n)$ counts all row-increasing arrays, before imposing any comparisons between different rows.

Theorem 2.1 (All orders at fixed height). *For every fixed integer $m \geq 1$ there is a uniquely determined sequence $c_{m,j} \in \mathbb{Q}$, with $c_{m,0} = 1$, such that, for every fixed $L \geq 0$,*

$$T_m(n) = M_m(n) \frac{J_m}{(4n)^d} \left(\sum_{j=0}^L \frac{c_{m,j}}{n^j} + O_{m,L}(n^{-L-1}) \right). \quad (2.3)$$

Every coefficient is given by the finite moment formula (5.9). In particular,

$$c_{m,1} = \frac{m(m^2 - 1)}{12}, \quad c_{m,2} = \frac{m^2(m^2 - 1)(m^2 + 2)}{288}. \quad (2.4)$$

Equivalently, there are rational $b_{m,j}$, $b_{m,0} = 1$, such that

$$T_m(n) = K_m m^{mn} n^{-\alpha_m} \left(\sum_{j=0}^L \frac{b_{m,j}}{n^j} + O_{m,L}(n^{-L-1}) \right), \quad (2.5)$$

where

$$b_{m,1} = \frac{(m^2 - 1)^2}{12m}, \quad (2.6)$$

$$b_{m,2} = \frac{(m^2 - 1)(m^6 + 3m^2 - 1)}{288m^2}. \quad (2.7)$$

Corollary 2.2 (A181199, with five corrections). *With $a(n) = T_5(n)$,*

$$a(n) = \frac{9 \cdot 5^{n+1/2}}{2^{17} \pi^2 n^{12}} \left(1 + \frac{48}{5n} + \frac{5233}{100n^2} + \frac{1028391}{5000n^3} + \frac{60248377}{100000n^4} + \frac{273939939}{250000n^5} + O(n^{-6}) \right). \quad (2.8)$$

Theorem 2.3 (Algebraicity dichotomy). *The ordinary generating function*

$$F_m(z) = \sum_{n \geq 0} T_m(n) z^n$$

is algebraic over $\mathbb{Q}(z)$ if and only if $m = 1$ or $m = 2$. The same classification holds for algebraicity over $\mathbb{C}(z)$.

Corollary 2.4 (A universal comparison with ordinary rectangles). *Let $f^{(n^m)}$ count ordinary standard Young tableaux of rectangular shape with m rows of length n . For fixed m ,*

$$\frac{T_m(n)}{f^{(n^m)}} = 2^{-m(m-1)} \left(1 + \frac{m(m^2 - 1)}{4n} + \frac{m^2(m^2 - 1)(m^2 - 2)}{32n^2} + O_m(n^{-3}) \right). \quad (2.9)$$

Thus the extra antidiagonal comparisons retain a positive limiting fraction $2^{-m(m-1)}$ of ordinary rectangular tableaux.

The constant is not obtained by treating the comparisons as independent. It comes from two shifted-hook denominators, one for each side of a threshold cut. Their product is the essential feature of the proof. The distributional counterpart is given in Section 9.

[write] Added 3 October 2026, batch 85: the comparison constant. The normalizer of Corollary 2.4 is the classical hook count of the rectangle, recorded in ProveIt as equation (3.19), label `t2:eq:tableaux`, of the volume *Combinatorial Transseries and Their Inverses* (Analysis/Transseries/docs/series-and-transseries/Combinatorial_Transseries_Inverses/, Section 3.5), which writes it $T_d(n)$ with d rows; the hook product displayed at the end of Section 6.4 here is the same formula with $d = m$. That volume's constants (3.20) (`t2:eq:tableaux-constants`) give $f^{(n^m)} \sim C m^{mn} n^\beta$ with $a = m \log m$, $\beta = -(m^2 - 1)/2 = -\alpha_m$ and $C = \sqrt{m} (2\pi)^{(1-m)/2} \prod_{j=0}^{m-1} j!$. Comparing with (2.2), $K_m = 2^{-m(m-1)} C$ exactly, since $4^d = 2^{m(m-1)}$: the two families share the rate m^m and the exponent $-\alpha_m$, and the leading term of (2.9) is the quotient of the two constants. The intake checked this identity symbolically for $m = 2, \dots, 7$, and checked that (2.4) and the rising-factorial expansion of $\prod_{j < m} (n + j)!/n!$ give the two printed corrections of (2.9) for the same m . Neither the formula nor the constant of the ordinary rectangle is new here; the new statement is the shifted count and its ratio.

3 Prefix states and the shifted hook product

3.1 The exact state space

Fill an array by inserting $1, 2, \dots, mn$ in order. At any intermediate stage let x_i be the number of filled cells in row i . Row increase means that the filled cells in each row are a prefix. The legal states satisfy

$$n \geq x_1 \geq x_2 \geq \dots \geq x_m \geq 0, \quad x_i = x_{i+1} \implies x_i \in \{0, n\}. \quad (3.1)$$

Indeed, placing a filled cell in row $i + 1$, column $j < n$, forces the cell in row i , column $j + 1$, to have already been filled. When the lower row has length n , column increase forces the upper row to be full as well. Conversely, these conditions make the filled set an order ideal of the array poset. They therefore characterize the prefix states exactly.

A transition increases x_i by one. For $i > 1$ it is allowed precisely when

$$x_{i-1} \geq x_i + 2 \quad \text{or} \quad (x_i = n - 1 \text{ and } x_{i-1} = n), \quad (3.2)$$

with $x_i < n$; the first row has only the latter upper bound. Empty rows can tie, and completed rows can tie. Forgetting either exception would change the sequence.

For a legal state $\lambda = (\lambda_1, \dots, \lambda_m)$ define its rotated complement by

$$\lambda^c = (n - \lambda_m, \dots, n - \lambda_1). \quad (3.3)$$

Rotation through 180 degrees, together with reversal of the numerical labels, preserves the comparisons. Thus suffix enumeration at λ is prefix enumeration at λ^c .

3.2 Interior states

Call a state *interior* when

$$n > \lambda_1 > \dots > \lambda_m > 0. \quad (3.4)$$

Every path ending there has avoided the completed-row boundary. After removing trailing zero parts, it is a chain of strict partitions obtained by adding one box at a time. Such chains are standard shifted tableaux. If g^λ denotes their number and $k = |\lambda|$, the shifted hook product is

$$g^\lambda = \frac{k!}{\prod_i \lambda_i!} \prod_{i < j} \frac{\lambda_i - \lambda_j}{\lambda_i + \lambda_j}. \quad (3.5)$$

This standard formula appears, for example, in Sun's account [9]. Appendix A proves it directly from the strict-partition recursion; no shifted-tableau enumeration result is needed without proof.

For an interior state both λ and λ^c are strict positive partitions. Consequently, the prefix and suffix counts are simultaneously available in product form. This is why cutting *in the middle*, rather than tracking the first completed row, is effective for asymptotics.

4 An exact random-threshold reduction

4.1 Order statistics and the cut

Take m independent rows, each consisting of n independent uniform random numbers in $(0, 1)$, sorted increasingly. Let $\mathcal{A}$ be the event that the comparisons between rows also hold. Every interleaving of these

sorted rows has the same probability, so

$$\mathbb{P}(\mathcal{A}) = \frac{T_m(n)}{M_m(n)}. \quad (4.1)$$

This is the order-statistics form of the order-polytope interpretation [9]; it also follows immediately by sorting all mn samples at once.

Fix $p \in (0, 1)$ and let X_i count the entries in row i below p . Before conditioning on $\mathcal{A}$, the variables X_i are independent $\text{Bin}(n, p)$. Conditional on these counts, the samples below p and above p are independent sorted uniforms in their respective intervals. When the count vector is an interior state λ , the conditional probability of satisfying all comparisons below the cut is

$$\frac{g^\lambda}{k! / \prod_i \lambda_i!}.$$

The analogous suffix probability uses λ^c . All comparisons across the cut are automatic because the prefix is an order ideal. Multiplying and using (3.5) gives

$$\mathbb{P}(\mathcal{A} \mid X = \lambda) = R_n(\lambda), \quad R_n(x) = \prod_{i < j} \frac{(x_i - x_j)^2}{(x_i + x_j)(2n - x_i - x_j)}. \quad (4.2)$$

The weight is symmetric in the coordinates. Equal interior coordinates give zero, and all distinct coordinates have exactly one decreasing order.

Theorem 4.1 (Interior sum and explicit boundary error). *Let $X_1, \dots, X_m$ be independent $\text{Bin}(n, p)$ variables and $\mathcal{I} = \{1 \leq X_i \leq n - 1 \text{ for all } i\}$. Then*

$$\frac{T_m(n)}{M_m(n)} = \frac{1}{m!} \mathbb{E} [\mathbf{1}_{\mathcal{I}} R_n(X)] + E_{m,n}(p), \quad (4.3)$$

where the error is nonnegative and satisfies

$$0 \leq E_{m,n}(p) \leq m(p^n + (1 - p)^n). \quad (4.4)$$

For $p = 1/2$ the expectation is the finite rational sum

$$I_m(n) = 2^{-mn} \sum_{n > x_1 > \dots > x_m > 0} \left(\prod_{i=1}^m \binom{n}{x_i} \right) \prod_{i < j} \frac{(x_i - x_j)^2}{(x_i + x_j)(2n - x_i - x_j)}, \quad (4.5)$$

and

$$0 \leq T_m(n) - M_m(n)I_m(n) \leq 2m 2^{-n} M_m(n). \quad (4.6)$$

Proof. The contribution to $\mathbb{P}(\mathcal{A})$ from interior count vectors is the sum of their binomial probabilities times (4.2). Symmetry converts this decreasing-vector sum into $1/m!$ times the unrestricted expectation on $\mathcal{I}$. The remaining event is $\mathcal{A} \cap \mathcal{I}^c$. It has nonnegative probability, at most the probability that some row has all its samples on one side of the cut. A union bound gives (4.4). The specialization is immediate. $\square$

Remark 4.2 (A useful pointwise bound). For $0 \leq x, y \leq n$, $|x - y| \leq x + y$ and $|x - y| \leq 2n - x - y$. Thus every factor of R_n lies in $[0, 1]$ whenever its denominator is nonzero. In particular, $0 \leq R_n \leq 1$ on $\mathcal{I}$. No uncontrolled singularity is hidden near the endpoints in (4.3).

The boundary estimate is an exact, nonasymptotic statement. It is deliberately simple and is not claimed sharp. Since the main probability is of order n^{-d} , this exponentially small additive error is negligible relative to every term of the inverse-power expansion.

[write] Added 5 October 2026, batch 98. The rate 2^{-n} of the bound at $p = 1/2$ is the true one, but the bound is not sharp in its constant, nor, for $m \geq 3$, in its power of n : the write's Proposition 18.3 in Part II shows $E_{m,n}(1/2) \sim 2^{1-n} 3^{1-m} J_{m-1}(4n)^{-\binom{m-1}{2}}$, against the bound $2m 2^{-n}$.

5 Gaussian reduction and an effective all-order theorem

5.1 The scaled integrand

Use $p = 1/2$ and put

$$Y_i = \frac{2X_i - n}{\sqrt{n}}, \quad \Delta(y) = \prod_{i < j} (y_i - y_j), \quad q_{ij}(y) = (y_i + y_j)^2. \quad (5.1)$$

The pair weight becomes exactly

$$R_n(X) = (4n)^{-d} \Delta(Y)^2 \prod_{i < j} \left(1 - \frac{q_{ij}(Y)}{4n}\right)^{-1} \quad \text{on } \mathcal{I}. \quad (5.2)$$

Let $h_r(q)$ be the complete homogeneous polynomial of degree r in the d variables q_{ij} , defined by

$$\prod_{i < j} (1 - u q_{ij})^{-1} = \sum_{r \geq 0} h_r(q) u^r. \quad (5.3)$$

In particular $h_0 = 1$. For $m = 1$, the empty product is one and $h_r = 0$ for $r > 0$.

5.2 Polynomial standardized moments

A variable Y_i has the same law as $n^{-1/2}(\varepsilon_1 + \cdots + \varepsilon_n)$, where the ε_j are independent symmetric signs. Its moment generating function is

$$\mathbb{E} e^{tY_i} = (\cosh(t/\sqrt{n}))^n. \quad (5.4)$$

For every fixed nonnegative integer k , its k th moment is exactly a polynomial $\mu_k(z)$ at $z = 1/n$. One convenient formal definition is

$$\sum_{k \geq 0} \mu_k(z) \frac{t^k}{k!} = \exp \left(\frac{1}{z} \log \cosh(t\sqrt{z}) \right). \quad (5.5)$$

The apparent singularity at $z = 0$ is removable in the formal expansion. Odd moments vanish, and

$$\begin{aligned} \mu_0 &= 1, & \mu_2 &= 1, & \mu_4 &= 3 - 2z, & (5.6) \\ \mu_6 &= 15 - 30z + 16z^2, & \mu_8 &= 105 - 420z + 588z^2 - 272z^3. & (5.7) \end{aligned}$$

For $k = 2s > 0$, the degree is at most $s - 1$. This follows either by grouping terms according to equal sign indices, or from the moment–cumulant formula: a partition into even blocks contributes $z^{s-\#\text{blocks}}$ times a product of sign cumulants.

Define a linear functional $\mathcal{E}_z : \mathbb{Q}[y_1, \dots, y_m] \rightarrow \mathbb{Q}[z]$ by

$$\mathcal{E}_z(y_1^{k_1} \cdots y_m^{k_m}) = \prod_{i=1}^m \mu_{k_i}(z). \quad (5.8)$$

Then $\mathcal{E}_{1/n}$ is the exact expectation for the independent variables Y_i . At $z = 0$, it is expectation under independent standard normal variables.

5.3 The coefficient formula

Proposition 5.1 (Finite formula for each coefficient). *For $j \geq 0$, the coefficient in (2.3) is*

$$c_{m,j} = \frac{1}{m!J_m} \sum_{r=0}^j \frac{1}{4^r} [z^{j-r}] \mathcal{E}_z(\Delta(y)^2 h_r(q(y))). \quad (5.9)$$

The formula uses finite polynomials and rational arithmetic. It does not require previously computed sequence terms or a guessed recurrence.

5.4 Uniform error control

We supply the estimates that justify the formal calculation. Let

$$C_n = \{|Y_i| \leq \sqrt{n}/2 \text{ for all } i\}.$$

Since $\cosh t \leq e^{t^2/2}$, Chernoff's inequality gives

$$\mathbb{P}(C_n^c) \leq 2me^{-n/8}. \quad (5.10)$$

The same moment generating function bound implies that the absolute moments of every fixed degree of the Y_i are bounded uniformly in n .

On C_n , each $q_{ij}(Y)/(4n)$ lies in $[0, 1/4]$. Taylor's theorem applied to the finite product in (5.2) therefore gives, for every fixed L ,

$$\prod_{i < j} \left(1 - \frac{q_{ij}(Y)}{4n}\right)^{-1} = \sum_{r=0}^L \frac{h_r(q(Y))}{(4n)^r} + \mathcal{R}_{L,n}(Y), \quad (5.11)$$

$$|\mathcal{R}_{L,n}(Y)| \leq C_{m,L} n^{-L-1} (1 + \|Y\|)^{2L+2}. \quad (5.12)$$

To see the second bound, differentiate the product $L + 1$ times with respect to $1/n$. Every differentiated term has total numerator degree $2L + 2$; all denominators are bounded below by fixed powers of $3/4$. After multiplication by $\Delta(Y)^2$, the expected remainder is still $O_{m,L}(n^{-L-1})$ by the uniform moment bounds.

We may replace polynomial expectations on C_n by unrestricted ones with exponentially small error. For example, Cauchy–Schwarz, (5.10), and a uniform moment bound control the difference for every fixed polynomial. For the original rational weight, use $R_n \leq 1$ from Remark 4.2. After restoring the factor $(4n)^d$, its discarded tail is only a fixed polynomial in n times an exponential. The endpoint error (4.4) behaves similarly. We have consequently proved

$$\frac{T_m(n)}{M_m(n)} = \frac{(4n)^{-d}}{m!} \left(\sum_{r=0}^L \frac{\mathcal{E}_{1/n}(\Delta^2 h_r)}{(4n)^r} + O_{m,L}(n^{-L-1}) \right). \quad (5.13)$$

This establishes (5.9) after collecting powers. It also proves that parity effects cannot occur in the algebraic sector: the moments are exact polynomials in $1/n$, on both parity subsequences.

5.5 The leading Gaussian integral

For independent standard normals $G_1, \dots, G_m$,

$$\mathbb{E}\Delta(G)^2 = m! \prod_{j=0}^{m-1} j! = m! J_m. \quad (5.14)$$

Here is an elementary verification. Replace the monomials in the Vandermonde determinant by the monic Hermite polynomials H_j , whose squared norms for standard normal measure are $j!$. The determinant is unchanged because this basis change is unit triangular. Expanding the square and using orthogonality leaves the $m!$ matching permutations, each contributing $\prod j!$.

Thus $c_{m,0} = 1$. Finally, Stirling's expansion gives

$$M_m(n) \sim \sqrt{m} (2\pi n)^{(1-m)/2} m^{mn}, \quad (5.15)$$

which proves the leading constant and exponent in (2.5). More explicitly, the complete conversion between the two normalizations is

$$\sum_{j \geq 0} b_{m,j} z^j = \left(\sum_{j \geq 0} c_{m,j} z^j \right) \exp \left(\sum_{r \geq 1} \frac{B_{2r}}{2r(2r-1)} (m^{1-2r} - m) z^{2r-1} \right), \quad (5.16)$$

where B_{2r} are Bernoulli numbers. All uses of this identity are formal, or truncated to a fixed finite order with Stirling's remainder estimate [12]. This completes the proof of Theorem 2.1, apart from the explicit evaluations in the next section.

6 Universal first and second corrections

The finite formula is already an all-order algorithm. To obtain short formulas valid at every height, it is better to keep the Vandermonde weight intact rather than expand it into individual monomials.

6.1 An exact binomial Vandermonde normalization

Define the normalized binomial ensemble by weighting the independent Y_i with $\Delta(Y)^2$. Its normalizing factor is

$$Z_m(n) = \mathbb{E}\Delta(Y)^2 = m! J_m \prod_{j=0}^{m-1} \frac{n^j}{n^j}. \quad (6.1)$$

This identity is valid for $n \geq m-1$ (and interpreted as zero when the support cannot accommodate m distinct points).

For completeness, let $X \sim \text{Bin}(n, 1/2)$ and define $K_j(X)$ by

$$(1+t)^X (1-t)^{n-X} = \sum_{j=0}^n K_j(X) t^j.$$

Multiplying two such generating functions and taking expectation gives $(1+st)^n$. Hence the K_j are orthogonal and $\mathbb{E}K_j^2 = \binom{n}{j}$. Their leading coefficient as polynomials in $Y = (2X - n)/\sqrt{n}$ is $n^{j/2}/j!$. The corresponding monic polynomials therefore have squared norms

$$h_j = j! n^{j/2}/n^j.$$

The same determinant argument as in (5.14) proves (6.1). These are the symmetric Krawtchouk polynomials; the standard orthogonal-polynomial framework is recorded in [13].

Put

$$A = \binom{m}{3}, \quad B = \frac{m(m-1)^2(m-2)}{12}. \quad (6.2)$$

Expanding the finite product in (6.1) gives

$$\frac{Z_m(n)}{m!J_m} = 1 - \frac{A}{n} + \frac{A^2 - B}{2n^2} + O_m(n^{-3}). \quad (6.3)$$

Indeed, $A = \sum_{j < m} \sum_{k < j} k$ and $B = \sum_{j < m} \sum_{k < j} k^2$.

6.2 The first pair statistic

Let $p_r = \sum_i Y_i^r$. The first complete homogeneous polynomial is

$$S_1 = h_1(q) = \sum_{i < j} (Y_i + Y_j)^2 = (m-2)p_2 + p_1^2. \quad (6.4)$$

Write $\langle \cdot \rangle_n$ for expectation in the normalized binomial Vandermonde ensemble. The monic polynomial recurrence is

$$yP_j(y) = P_{j+1}(y) + \beta_j P_{j-1}(y), \quad \beta_j = j \left(1 - \frac{j-1}{n}\right).$$

The diagonal coefficient is zero by symmetry. The norm quotient proves the displayed value of β_j .

We need the standard finite-projection identities

$$\langle p_2 \rangle_n = \beta_m + 2 \sum_{j=1}^{m-1} \beta_j = m^2 - \frac{m(m-1)(2m-1)}{3n}, \quad (6.5)$$

$$\langle p_1^2 \rangle_n = \beta_m = m - \frac{m(m-1)}{n}. \quad (6.6)$$

They can be seen directly from the determinant ensemble: multiplication by y is tridiagonal in the orthonormal basis, with off-diagonal entries $\sqrt{\beta_j}$. The mean of a linear statistic is its multiplication operator's trace on degrees $0, \dots, m-1$. Its variance is the squared norm of the part crossing that projection. For p_1 only the edge from degree $m-1$ to degree m crosses, giving β_m . These trace rules also follow by expanding $\mathbb{E} \prod_i (1 + tf(Y_i))$ as a determinant of the moment matrix and taking the first two derivatives at $t = 0$.

Consequently,

$$\langle S_1 \rangle_n = s_{10} + \frac{s_{11}}{n}, \quad s_{10} = m(m-1)^2, \quad s_{11} = -\frac{m(m-1)(2m^2 - 5m + 5)}{3}. \quad (6.7)$$

6.3 The second pair statistic in the Gaussian limit

Let $\langle \cdot \rangle_0$ denote the Gaussian Vandermonde ensemble. Since

$$h_2(q) = \frac{1}{2} \left(S_1^2 + \sum_{i < j} (Y_i + Y_j)^4 \right),$$

and

$$\sum_{i < j} (Y_i + Y_j)^4 = (m - 8)p_4 + 4p_1p_3 + 3p_2^2,$$

we need only a few Gaussian moments:

$$\langle p_2^2 \rangle_0 = m^2(m^2 + 2), \quad \langle p_2p_1^2 \rangle_0 = m(m^2 + 2), \quad (6.8)$$

$$\langle p_1^4 \rangle_0 = 3m^2, \quad \langle p_4 \rangle_0 = 2m^3 + m, \quad \langle p_1p_3 \rangle_0 = 3m^2. \quad (6.9)$$

There is no need to invoke random-matrix integration to establish these. Scaling and translation of the Gaussian integral give

$$\langle e^{tp_2 + sp_1} \rangle_0 = (1 - 2t)^{-m^2/2} \exp\left(\frac{ms^2}{2(1 - 2t)}\right). \quad (6.10)$$

This proves the first three identities by differentiation. Translation integration by parts, using $\Delta(y + c\mathbf{1}) = \Delta(y)$, gives $\langle p_1p_3 \rangle_0 = 3\langle p_2 \rangle_0 = 3m^2$. Finally, in the Hermite multiplication matrix, the degree- j diagonal entry of the fourth power is $6j^2 + 6j + 3$; summing over $0 \leq j < m$ yields $2m^3 + m$. Thus

$$H_{20} := \langle h_2(q) \rangle_0 = \frac{m(m - 1)(m^4 - 3m^3 + 10m^2 - 18m + 16)}{2}. \quad (6.11)$$

6.4 Collecting the terms

Combining (6.3), (6.7), and (6.11),

$$c_{m,1} = -A + \frac{s_{10}}{4}, \quad (6.12)$$

$$c_{m,2} = \frac{A^2 - B}{2} + \frac{-As_{10} + s_{11}}{4} + \frac{H_{20}}{16}. \quad (6.13)$$

Straight polynomial simplification gives (2.4). The first Stirling logarithm coefficient is $u_1 = (m^{-1} - m)/12$, so $b_{m,1} = c_{m,1} + u_1$ and $b_{m,2} = c_{m,2} + u_1c_{m,1} + u_1^2/2$. This proves (2.7).

The ordinary rectangular hook product is

$$f^{(n^m)} = M_m(n)J_m \prod_{j=0}^{m-1} \frac{n!}{(n+j)!}.$$

Its finite rising-factorial expansion, combined with (2.4), proves (2.9).

7 OEIS specializations and numerical checks

7.1 Low-height controls

For $m = 1$, $T_1(n) = 1$. For $m = 2$, the usual strict-ballot argument gives $T_2(n) = C_{n-1}$ for $n \geq 1$; equivalently,

$$T_2(n) = \binom{2n}{n} \frac{1}{2(2n-1)} = \binom{2n}{n} \frac{1}{4n} \sum_{j \geq 0} \frac{1}{(2n)^j}. \quad (7.1)$$

Thus $c_{2,j} = 2^{-j}$ at every order. This is a useful exact control of both the normalization and the coefficient generator.

For height three, the theorem recovers the existing A181197 leading term [2] and gives

$$T_3(n) = \frac{\sqrt{3} 27^n}{64\pi n^4} \left(1 + \frac{16}{9n} + \frac{755}{324n^2} + O(n^{-3}) \right). \quad (7.2)$$

For height four, corresponding to A181198,

$$T_4(n) = \frac{3 \, 256^n}{1024\sqrt{2} \pi^{3/2} n^{15/2}} \left(1 + \frac{75}{16n} + \frac{6905}{512n^2} + O(n^{-3}) \right). \quad (7.3)$$

The leading constants are consequences of the general theorem, not inputs fitted from these sequences.

7.2 Five corrections in multinomial normalization

Table 1 reports exact output of (5.9). Only the compact multinomial normalization is used in this table. The archive also supplies the corresponding $b_{m,j}$.

Table 1: Coefficients $c_{m,j}$ in (2.3); $c_{m,0} = 1$.

m	$c_{m,1}$	$c_{m,2}$	$c_{m,3}$	$c_{m,4}$	$c_{m,5}$
2	1/2	1/4	1/8	1/16	1/32
3	2	11/4	23/8	45/32	-57/16
4	5	15	33	789/16	93/16
5	10	225/4	1819/8	22045/32	1353
6	35/2	665/4	8939/8	92351/16	731621/32

7.3 How much the corrections improve A181199

For $m = 5$, let $L_5(n) = K_5 5^5 n^{-12}$ and $A_{5,L}(n) = L_5(n) \sum_{j=0}^L b_{5,j} n^{-j}$. Table 2 shows the signed relative error $A_{5,L}(n)/T_5(n) - 1$. Its values are numerical diagnostics, not rigorous interval enclosures. The counts at $n = 10, 20, 40$ were independently regenerated by the row-state dynamic program and compared with OEIS; $n = 80$ is an OEIS reference value, not regenerated in this run [6].

Table 2: A181199: signed relative errors of successive truncations.

n	$L = 0$	$L = 1$	$L = 2$	$L = 5$
10	$-6.37190 \cdot 10^{-1}$	$-2.88893 \cdot 10^{-1}$	$-9.90345 \cdot 10^{-2}$	$1.42184 \cdot 10^{-3}$
20	$-3.90471 \cdot 10^{-1}$	$-9.78967 \cdot 10^{-2}$	$-1.81550 \cdot 10^{-2}$	$1.97547 \cdot 10^{-5}$
40	$-2.16403 \cdot 10^{-1}$	$-2.83393 \cdot 10^{-2}$	$-2.71080 \cdot 10^{-3}$	$2.67266 \cdot 10^{-7}$
80	$-1.13941 \cdot 10^{-1}$	$-7.61418 \cdot 10^{-3}$	$-3.69269 \cdot 10^{-4}$	$3.76974 \cdot 10^{-9}$

The leading term alone converges rather slowly at height five because $b_{5,1} = 9.6$. This explains why the constant is difficult to recognize reliably from modestly sized data without correction terms. The proof, however, does not use the numerical agreement.

8 A complete algebraicity classification

We now prove Theorem 2.3. The argument uses the classical Newton–Puiseux property: an algebraic function has a convergent fractional power expansion at a finite point, with algebraic coefficients when both the defining polynomial and the expansion point are algebraic. It has no logarithmic leading singularity. This is standard algebraic-function background; see the algebraic-function treatment in [11].

Put $R_m = m^{-m}$ and rescale

$$G_m(z) = F_m(R_m z) = \sum_{n \geq 0} v_n z^n, \quad v_n = R_m^n T_m(n).$$

For $m \geq 2$, Theorem 2.1 gives

$$v_n = K_m n^{-\alpha_m} (1 + O(n^{-1})). \quad (8.1)$$

In particular the radius of convergence of G_m is exactly one.

8.1 Odd heights: a logarithmic obstruction

Suppose $m \geq 3$ is odd. Then α_m is a positive integer. Set $q = \alpha_m - 1$. The coefficient of z^k in $G_m^{(q)}(z)$ is

$$\frac{(k+q)!}{k!} v_{k+q} = \frac{K_m}{k} + O(k^{-2}) \quad (k \rightarrow \infty).$$

After a harmless adjustment of finitely many terms, subtraction of $K_m \sum_{k \geq 1} z^k/k$ leaves an absolutely summable coefficient sequence. Therefore, along the real interval $z \uparrow 1$,

$$G_m^{(q)}(z) = K_m \log \frac{1}{1-z} + O(1). \quad (8.2)$$

A nonzero logarithmic divergence is incompatible with a Puiseux expansion. If F_m were algebraic, so would be its rescaling and every derivative; (8.2) is a contradiction.

8.2 Even heights: a transcendental-amplitude obstruction

Suppose $m \geq 4$ is even. Now $q = \alpha_m - 1/2 = (m^2 - 2)/2$ is an integer. The derivative coefficients satisfy

$$[z^k] G_m^{(q)}(z) = K_m k^{-1/2} + O(k^{-3/2}).$$

Since $[z^k](1-z)^{-1/2} = 1/\sqrt{\pi k} + O(k^{-3/2})$, coefficient subtraction again leaves an absolutely summable series. Hence

$$G_m^{(q)}(z) = K_m \sqrt{\pi} (1-z)^{-1/2} + O(1), \quad z \uparrow 1. \quad (8.3)$$

If the function were algebraic over $\mathbb{Q}(z)$, the leading coefficient in this Puiseux expansion at the algebraic point $z = 1$ would be algebraic. But

$$K_m \sqrt{\pi} = \frac{\sqrt{m} J_m}{4^d 2^{(m-1)/2}} \pi^{-(m-2)/2} \quad (8.4)$$

is a nonzero algebraic multiple of a nonzero integer power of π^{-1} . It is transcendental: otherwise that power of π , and hence π itself, would be algebraic. This contradiction uses only the classical transcendence of π .

8.3 The two algebraic cases and descent of constants

The exceptional cases are explicit:

$$F_1(z) = \frac{1}{1-z}, \quad F_2(z) = 1 + zC(z), \quad C(z) = \frac{1 - \sqrt{1-4z}}{2z}.$$

Thus $m = 1, 2$ are algebraic.

Finally, a rational-coefficient power series algebraic over $\mathbb{C}(z)$ is algebraic over $\mathbb{Q}(z)$. Indeed, fix finite degree bounds for a nonzero polynomial relation. Equating all coefficients gives a homogeneous linear system with rational entries in finitely many unknown polynomial coefficients. The descending chain of solution spaces stabilizes after finitely many independent equations. A nonzero complex solution therefore implies a nonzero rational solution. This proves the asserted classification over either field.

Remark 8.1. Nonalgebraicity does not imply non-D-finiteness. Logarithmic singularities, and half-integer singularities with transcendental connection constants, are compatible with D-finite functions. The guessed recurrences for heights four and five remain separate questions.

[write] **Added 5 October 2026, batch 98.** F_m is in fact D-finite for every fixed m : it is the diagonal of a rational power series over $\mathbb{Q}$ (Part II, Theorem 14.1, through the multiple-binomial-sum theorem of Bostan, Lairez and Salvy). With Theorem 2.3, F_m is algebraic for $m \leq 2$ and transcendental but D-finite for $m \geq 3$. That proof is an existence proof; it produces no operator, and the guessed recurrences for heights four and five remain unproved.

9 Gaussian laws for intermediate shapes

The same reduction describes typical partially filled arrays, not only their total number. We give both a randomized numerical cut and a fixed number of inserted labels; these are different probability models.

9.1 A cut at a fixed numerical level

Choose a tableau uniformly among the $T_m(n)$ arrays. Independently choose mn uniform random numbers in $(0, 1)$, sort them, and place them in the array in increasing order of its integer labels. This produces the uniform order-statistics model conditioned on $\mathcal{A}$. For a fixed $p \in (0, 1)$ let X_i be the number of row- i entries below p and set

$$Z_i = \frac{X_i - np}{\sqrt{np(1-p)}}. \quad (9.1)$$

Theorem 9.1 (Gaussian Vandermonde limit). *For fixed m and fixed $p \in (0, 1)$, the decreasing vector Z converges in distribution to the probability density*

$$\frac{e^{-\|z\|^2/2} \Delta(z)^2}{(2\pi)^{m/2} J_m}, \quad z_1 > z_2 > \cdots > z_m, \quad (9.2)$$

with respect to m -dimensional Lebesgue measure. Polynomial moments of fixed degree converge as well.

Proof. For interior decreasing x , the exact conditional mass is

$$\frac{\prod_i \mathbb{P}(\text{Bin}(n, p) = x_i) R_n(x)}{T_m(n)/M_m(n)}.$$

In bounded sets of standardized coordinates, the local central limit estimate for the binomial mass follows directly from Stirling's formula. Moreover,

$$x_i - x_j = \sqrt{np(1-p)}(z_i - z_j),$$

whereas the two denominator factors in (4.2) are asymptotic to $2np$ and $2n(1-p)$. Thus

$$R_n(x) = (4n)^{-d} \Delta(z)^2 (1 + o(1))$$

uniformly on every bounded set. Divide by $T_m(n)/M_m(n) \sim J_m(4n)^{-d}$ and use the lattice mesh $1/\sqrt{np(1-p)}$ in every coordinate. The Riemann sums converge to (9.2).

To justify passage beyond bounded sets, first restrict every x_i/n to a fixed compact interval inside $(0, 1)$ containing p . There the pair denominators are bounded below by a constant times n^2 , and binomial local bounds give a Gaussian upper bound times a fixed polynomial in the standardized coordinates. This is integrable, with every polynomial moment. Outside that macroscopic interval, Chernoff bounds and $R_n \leq 1$ give an exponentially small contribution even after division by the order- n^{-d} conditioning probability. The boundary term satisfies the same conclusion by (4.4). This proves tightness and moment convergence. Normalization of the limit is (5.14), with the factor $m!$ removed on the decreasing chamber. $\square$

The density (9.2) is the ordered eigenvalue density of the Gaussian unitary ensemble in the normalization with Gaussian factor $e^{-\sum z_i^2/2}$. This terminology is optional: the theorem and its normalization have already been specified by an explicit density.

9.2 A deterministic fraction of the labels

Let $\Lambda_i(k)$ be the number of cells in row i occupied by labels $1, \dots, k$ of a uniformly random tableau. Fix $p \in (0, 1)$ and take $k = \lfloor mnp \rfloor$.

Theorem 9.2 (Trace-zero limit at deterministic time). *The vector*

$$\left(\frac{\Lambda_i(k) - np}{\sqrt{np(1-p)}} \right)_{i=1}^m$$

converges to the density

$$\frac{e^{-\|z\|^2/2} \Delta(z)^2}{(2\pi)^{(m-1)/2} J_m} \quad (9.3)$$

on $\{z_1 > \dots > z_m, \sum_i z_i = 0\}$, with respect to Euclidean $(m-1)$ -dimensional measure on the trace-zero hyperplane.

Proof. In the preceding randomized-cut construction, the total number K of samples below p is $\text{Bin}(mn, p)$ and is independent of the integer tableau. Conditioning on $K = k$ therefore gives precisely $\Lambda(k)$. For an interior vector x of total size k , divide the mass used in the preceding proof by $\mathbb{P}(K = k)$. The latter is asymptotic to $(2\pi mnp(1-p))^{-1/2}$. The lattice $\{x \in \mathbb{Z}^m : \sum x_i = k\}$ has Euclidean covolume $\sqrt{m}$. In standardized coordinates its cell volume is $\sqrt{m} (np(1-p))^{-(m-1)/2}$. These factors give exactly the density (9.3) under the local binomial estimate.

For domination, use the same compact macroscopic region as before. The product of local binomial upper bounds on the hyperplane, after the additional conditioning, is bounded by a fixed polynomial times a Gaussian in the standardized coordinates, times the hyperplane lattice-cell volume. Its tails are uniformly

summable. Outside the macroscopic region, the extra factor $1/\mathbb{P}(K = k) = O(\sqrt{n})$ cannot overcome an exponential Chernoff bound. This proves convergence of the hyperplane Riemann sums.

Finally, Δ is invariant under common translation. Orthogonally splitting $\mathbb{R}^m$ into the trace-zero hyperplane and the unit trace direction shows that the full Gaussian Vandermonde integral is $\sqrt{2\pi}$ times the trace-zero integral. Equation (5.14) then verifies the normalization in (9.3). $\square$

This theorem concerns one fixed macroscopic time. It is not a claim of process convergence to a noncolliding Brownian bridge; proving tightness in a path space and identifying the joint time laws require additional work.

10 Ratios, eventual log-convexity, and inversion

10.1 Consecutive ratios

Write $\alpha = \alpha_m$ and $b_1 = b_{m,1}$. For $m \geq 2$, the all-order expansion gives

$$\frac{T_m(n+1)}{T_m(n)} = m^m \left(1 - \frac{\alpha}{n} + \frac{\alpha(\alpha+1)/2 - b_1}{n^2} + O_m(n^{-3}) \right). \quad (10.1)$$

Hence $T_m(n)$ is eventually strictly increasing. Expanding $\log T_m(n)$ at $n-1, n, n+1$ with enough terms also gives

$$\log T_m(n+1) - 2 \log T_m(n) + \log T_m(n-1) = \frac{\alpha}{n^2} + O_m(n^{-3}) > 0 \quad (10.2)$$

for sufficiently large n . Thus it is eventually strictly log-convex. These statements do not identify the smallest valid n .

10.2 Inverting the asymptotic profile

Put $\lambda = m \log m$, $K = K_m$, and $L = \log(y/K)$. The large solution of

$$\lambda x - \alpha \log x = L$$

is

$$x_0 = -\frac{\alpha}{\lambda} W_{-1} \left(-\frac{\lambda}{\alpha} e^{-L/\alpha} \right), \quad (10.3)$$

where W_{-1} is the real branch taking values at most -1 [14]. The branch choice matters: the principal branch describes the small, irrelevant solution. In elementary logarithmic form,

$$x_0 = \frac{L + \alpha \log L - \alpha \log \lambda}{\lambda} + O_m \left(\frac{\log L}{L} \right). \quad (10.4)$$

Let $\ell_2 = b_{m,2} - b_{m,1}^2/2$. Formal inversion of the continuous asymptotic profile gives

$$x = x_0 - \frac{b_{m,1}}{\lambda x_0} - \frac{\alpha b_{m,1}/\lambda^2 + \ell_2/\lambda}{x_0^2} + O_m(x_0^{-3}). \quad (10.5)$$

To verify it, insert $x = x_0 + \delta$ into $\lambda x - \alpha \log x + b_{m,1}/x + \ell_2/x^2 = L$ and compare the coefficients of x_0^{-1} and x_0^{-2} . Along the exact values $y = T_m(n)$, the approximation in (10.5) returns $n + O_m(n^{-3})$. The higher inverse

terms are obtained by the same recursive substitution. This inversion technique is standard and also used elsewhere in ProveIt; its application here is a consequence of the new expansion, not a new inversion method.

For an arbitrary target y , the least integer index attaining it is a staircase function. An asymptotic estimate alone must not be rounded and advertised as an exact threshold certificate. Near an integer transition, explicit upper and lower bounds for the neighboring sequence values are required. The finite interior sum and its boundary error (4.6) supply one possible, though not always computationally efficient, route to such bounds.

[write] Added 3 October 2026, batch 85: the inversion is an instance. The inversion (10.3) is a case of Theorem J.23 (p0:thm:lambert-core, “Exact solution of the dominant block”) of the canonical transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`, which solves $aX + b \log X = L$ by $X = (b/a) W_k((a/b)e^{L/b})$. Here $a = \lambda = m \log m > 0$ and $b = -\alpha_m < 0$ for $m \geq 2$, and that theorem’s branch rule for $b < 0$ gives exactly (10.3): the large root is the W_{-1} root, and it exists precisely when $L > L_c = \alpha_m(1 - \log(\alpha_m/\lambda))$ (for $m = 5$, $K_5 e^{L_c} \approx 0.021$, so every $y \geq 1$ qualifies). The text’s own sentence that the technique “is standard and also used elsewhere in ProveIt” is accurate; nothing in this section is a new inversion result. The real substitution, branch rules, threshold and slope of Theorem J.23 are formalized in `Analysis/FabiusFunction/Lean/FabiusFunction/LinLogCoreInversion.lean` (the volume’s register rates the theorem “Partial”: the asymptotic and general complex-branch clauses are not formalized); that formalization concerns the general equation, and nothing specific to shifted rectangles is formalized. The caution about integer thresholds in the preceding paragraph is the separation condition of the volume’s staircase theorem, Theorem J.43 (p0:thm:staircase). The path-forest report cited in Section 1.3 inverts its sequences by the same apparatus (its `spf:thm:inverse-first` and `spf:thm:inverse-all`).

11 Reproducibility and computational scope

11.1 Exact coefficient generation

The program `code/coefficients.py` implements (5.9) using Python’s exact rational arithmetic. It avoids expanding the squared Vandermonde in full. For any symmetric polynomial P and any product measure with identical one-variable moments,

$$\mathcal{E}_z(\Delta^2 P) = m! \sum_{\sigma \in S_m} \text{sgn}(\sigma) \mathcal{E}_z \left(P \prod_{i=0}^{m-1} y_{i+1}^{i+\sigma(i)} \right). \quad (11.1)$$

Expand the two Vandermonde determinants, permute the integration variables in one expansion, and use symmetry of P to prove this identity. Only $m!$ permutations remain. Equal sorted exponent profiles are grouped before evaluating products of moment polynomials.

The h_r are generated from (5.3); the standardized moments are generated from sign cumulants. The Stirling conversion (5.16) is also done with rational numbers. The method is practical for modest fixed m and moderate order. Its permutation factor makes it unsuitable as presently implemented for large m ; no claim of polynomial complexity in m is made.

11.2 Independent enumeration and exact finite identities

The program `code/tableaux.py` supplies two enumerators: a row-state dynamic program using (3.2), and an independent bitmask linear-extension enumerator using the cell-level predecessor relation. The latter is intentionally restricted to small arrays. Neither uses a conjectured recurrence.

The verification run recorded in the archive passed 471 checks in the following suites:

Suite	Checks
Bitmask-poset versus row-state enumeration	27
Interior prefix counts versus the shifted hook product	238
Threshold split, rational interior sum, and boundary inequality	120
Exact Catalan control, $n = 1, \dots, 30$	30
Selected A181198 values, independently regenerated	12
Selected A181199 values, independently regenerated	12
Universal corrections versus the coefficient engine, $m = 1, \dots, 6$	30
Standardized binomial-moment controls	2

The independently regenerated OEIS comparisons use $n = 1, \dots, 10, 20, 40$ for each of heights four and five [5, 6]. The threshold identity is checked for $1 \leq m \leq 5$ and $1 \leq n \leq 8$. These are finite corroborations of the proofs, not substitutes for them.

11.3 Commands and data provenance

From the archive’s top-level directory:

```
python3 code/verify.py
python3 code/coefficients.py --height 5 --order 5
python3 code/numerics.py
bash build.sh
```

The first two commands use the Python standard library only. The numerical diagnostics additionally use `mpmath`; they were run with 90 decimal digits. The PDF build requires a conventional LaTeX installation with the packages listed in the preamble.

The reference integers in `data/oeis_selected.json` were transcribed from the entry pages and b-files inspected during the source audit. They are not outputs of the coefficient engine. Counts regenerated independently are saved separately in `data/exact_regenerated.json`. Values at $n = 80$ appear only as reference inputs to numerical diagnostics. Full third-party articles are not redistributed in this archive.

[write] Added 3 October 2026, batch 85: the shipped layout. “The archive” in this section is the delivered package. In ProvelT its files keep the subdirectories `code/` and `data/`, except four, which moved as follows:

Delivered as	Shipped as
<code>build.sh</code>	<code>code/build.sh</code>
<code>requirements-numerics.txt</code>	<code>data/requirements-numerics.txt</code>
<code>notes/SOURCES_AND_STATUS.md</code>	<code>notes-SOURCES_AND_STATUS.md</code>
<code>notes/PROPOSED_OEIS_ADDITIONS.txt</code>	<code>data/notes-PROPOSED_OEIS_ADDITIONS.txt</code>

`code/build.sh` changes to its own directory, so in its shipped place it no longer finds `article.tex`. The delivered PDF and checksum list are not shipped; they survive in the arrival commit’s archive. `code/verify.py` and `code/numerics.py` write their outputs into the report’s own `data/` directory, overwriting the shipped records, so they must be rerun on a copy; the README of the report gives the commands. At intake (3 October 2026, Python 3.14.4 on Windows) `code/verify.py` passed all 471 checks on a copy, and every regenerated data file agreed with the shipped one after removal of carriage returns, except the timing field of `data/verification.json`; `code/coefficients.py` and `code/numerics.py` (`mpmath` 1.3.0) reproduced `data/coefficients_m5.json` and `data/numerics.csv`.

`data/oeis_selected.json` holds terms of A181198 and A181199 copied from the OEIS entries and b-files (Koutschan, with earlier terms by Heinz); OEIS content is published by The OEIS Foundation Inc. under the Creative

Commons Attribution-ShareAlike 4.0 licence, so this file is third-party data under that licence, not MIT-0. The draft OEIS text in `data/notes-PROPOSED_OEIS_ADDITIONS.txt` is marked “DRAFT ONLY — NOT SUBMITTED” and stays so: nothing was submitted to OEIS by the package’s author or by the intake, and its own instruction not to mark either conjectured recurrence as proved stands.

12 Further research questions and proposed approaches

The following questions are not presented as solved. They are chosen to continue the present line of work rather than to restate the resolved leading-term conjecture.

Research question 1 (Prove the specific height-four and height-five recurrences). Can the exact enumeration be transformed into a finite hypergeometric sum or a Pfaffian identity that admits a compact creative-telelescoping certificate for the recurrences in A181198 and A181199?

The random-threshold expression is exact only after adding its boundary part. For a recurrence proof that part cannot be discarded. A useful first step is an exact decomposition according to the number of full and empty rows at the cut, followed by shifted-skew-tableau formulas for those boundary states. A proof should compare the resulting operator with the published guessed operator, not merely generate more confirming terms.

Research question 2 (Polynomial dependence of every correction on height). For each fixed j , is $c_{m,j}$ a polynomial in m of degree at most $3j$? Can its factorization and denominator structure be described uniformly?

The formulas for $j = 1, 2$ support this degree scale. The question beyond those orders is not proved here. A promising approach is to reorganize (5.9) as Gaussian matrix moments or as connected cumulant diagrams, in which the powers of m count index loops. This could replace factorial-in- m computation with symbolic fixed-order formulas.

Research question 3 (A growing-height transition). For what growth rates $m = m(n)$ does the leading fixed-height formula remain relatively accurate? Is the natural first transition governed by m^3/n ?

The coefficient $c_{m,1} \sim m^3/12$ suggests this parameter in multinomial normalization. It does not establish a uniform regime. A proof would need uniform Vandermonde moment bounds and a growing-dimension tail analysis. For m^3/n of order one, connected logarithmic corrections may require resummation rather than ordinary fixed-order truncation.

Research question 4 (Exponentially small sectors and sharp boundary asymptotics). What is the true leading asymptotic of the discarded boundary term $E_{m,n}(1/2)$, and how does it combine with large deviations of the interior expectation?

The bound $2m2^{-n}$ is sufficient for every inverse power but is not an asymptotic formula. Configurations with a row empty or full give a natural finite hierarchy of lower-dimensional boundary strata. Computing their constants could produce a genuine transseries beyond the algebraic sector, and could clarify the signs and optimal truncation of the coefficients.

[write] Added 5 October 2026, batch 98: **Question 4 re-scoped.** Part II does not answer this question, although its introduction counts the non-sharp boundary bound among “the two gaps addressed here”. Its rare count $R_m(n)$ is the part of the boundary event in which the first row is completed before the last row starts: it satisfies $R_m(n) \leq M_m(n)E_{m,n}(1/2)$, and the ratio grows like 2^n times a power of n (Remark 14.4). The leading asymptotic of $E_{m,n}(1/2)$ itself is the write’s

Proposition 18.3: exactly, $E_{m,n}(1/2)$ is twice the stratum with one boundary row (the last row empty at the cut, or, by symmetry, the first row full) plus $O(4^{-n})$, and

$$E_{m,n}(1/2) \sim 2^{1-n} 3^{1-m} J_{m-1}(4n)^{-\binom{m-1}{2}}.$$

That proposition is unrefereed and checked numerically only on finite ranges. Still open: an all-order expansion of the boundary term (the one-row stratum is an interior sum with $m - 1$ rows and an extra factor $x_i/(2n - x_i)$ per row, so the method of Section 5 should apply, but this has not been carried out); the constants of the deeper strata, with two or more rows empty or full at the cut, and of the rare sectors inside them (compare Research question 10 of Part II); their combination with large deviations of the interior expectation; and any transseries reading.

Research question 5 (A full path-process limit). Do the centered row-length processes of a uniform tableau converge to a trace-zero noncolliding Gaussian bridge on compact time intervals inside $(0, 1)$? What are the entrance and exit scalings near the endpoints?

Theorem 9.2 establishes one-time marginals only. A proof of process convergence would require multi-cut enumeration, control of increments, and tightness. The complement symmetry provides a useful constraint on any proposed entrance/exit description.

Research question 6 (Certified asymptotic enclosures and integer thresholds). Can the moment expansion be packaged into efficient explicit upper and lower bounds with practical constants, avoiding the $\binom{n-1}{m}$ -term interior sum?

On a central region the pair interaction has a positive geometric expansion. Combining its finite partial sums with concentration bounds and exact moment determinants could give certified intervals. These would turn the inverse approximations into exact index decisions away from a quantified transition zone.

Research question 7 (D-finiteness versus nonalgebraicity at arbitrary height). Is $F_m(z)$ D-finite for every fixed m , and how does the minimal annihilating order grow with m ?

Theorem 2.3 settles the algebraic/nonalgebraic distinction, not the D-finite/non-D-finite one. An exact finite-dimensional differential system in the fixed-height regime would be a stronger structural result. The opposite-orientation constant-width transfer matrix does not answer this question.

[write] **Added 5 October 2026, batch 98: Question 7 answered in part.** The first half is answered: for every fixed m , F_m is the diagonal of a rational power series over $\mathbb{Q}$, hence D-finite, and $(T_m(n))$ is P-recursive (Part II, Theorem 14.1: the exact sum of Theorem 16.1 over the $2m$ row births and completions, a finite sum of products of reflection determinants, is a multiple binomial sum in the sense of Bostan, Lairez and Salvy). The second half, how the minimal annihilating order (and degree) grows with m , remains open; it is Research question 11 of Part II. The existence proof supplies no operator, so the guessed recurrences of heights four and five (Research question 1) remain unproved.

13 Conclusion

A shifted rectangle admits a particularly effective asymptotic cut: both sides of an interior threshold state are ordinary strict shifted shapes. The two shifted hook products turn the count into a binomial Vandermonde expectation, with an explicitly controlled boundary error. This gives a proof of the A181199 leading constant independent of all guessed recurrences, an effective all-order expansion for every fixed height, and universal correction polynomials through second order.

The consequences go beyond adding coefficients to an OEIS entry. The same constant yields a complete algebraicity dichotomy for the fixed-height generating functions, while the same exact weight identifies Gaussian laws for typical intermediate shapes. The remaining recurrence, uniformity, and exponentially small questions are separate, concrete targets with clear starting points.

Part II

Rational diagonals and rare boundary phases in shifted rectangles

D-finiteness at every fixed height and a sharp exponential refinement for OEIS A181199

Prepared for Vladimir Reshetnikov, October 4, 2026

For every fixed height m , the generating function of the shifted-rectangle counts $T_m(n)$ is a diagonal of a rational function, hence D-finite. For $m \geq 2$, let $R_m(n)$ count arrays in which row 1 finishes before row m starts. Then

$$R_m(n) \sim \sqrt{\frac{m}{2}} (2\pi)^{-(m-2)/2} \frac{\prod_{j=0}^{m-3} j!}{4^{\binom{m-2}{2}} 9^{m-2}} \left(\frac{m^m}{4}\right)^n n^{-(m-2)^2/2}.$$

Both statements hold for all fixed heights, not only the tabulated cases.

Abstract. The fixed-height shifted-rectangle family combines strict inequalities between active rows with exceptional ties at empty and completed rows. We separate these boundary events from the intervening walks. The resulting finite sum of reflection determinants is a multiple binomial sum, proving rational diagonality and D-finiteness at every fixed height and resolving an explicit question in the preceding ProveIt report. We also count the rare failure of a common active phase. A first-completion cut gives a shifted-hook sum with an exponentially smaller remainder. A beta-binomial mixture then yields its sharp exponential rate, leading constant, effective all-order expansion, and a universal first correction. We compute three corrections at height five, describe the conditional Gaussian shape, and invert the rare-event profile. Exact verification programs and a source and status audit accompany the manuscript.

Status. AI-assisted, unrefereed research manuscript. The stated results have mathematical proofs below, but have not been independently peer reviewed or proof-assistant verified. Historical priority is not certified. The particular conjectured recurrences and finite-sum identities for A181198 and A181199 are *not* proved here.

[write] Added 5 October 2026, batch 98: provenance of Part II. Part II is a single manuscript, printed in full: manuscript 02 of batch 98 of ProveIt’s incoming-reports intake, *Rational diagonals and rare boundary phases in shifted rectangles*, subtitled “D-finiteness at every fixed height and a sharp exponential refinement for OEIS A181199” and dated October 4, 2026 (a 1,656-line source with 97 labels and a 24-page PDF). Its title page reads “Prepared for Vladimir Reshetnikov”; its PDF metadata name “Research report prepared for Vladimir Reshetnikov”. No human author and no tool are named. It arrived as the archive `Shifted_Rectangle_Boundary_Research.zip` (390,943 bytes, 22 files under a doubled wrapper directory of the same name) in commit 2172df76a and was placed in commit 0fba5167f (batch 98B) as an addition to this report. It pins the repository at commit db20eb37982abc0f163c6d308d392f5ac4c9bc62 (4 October 2026; Section 14). This report was not changed between the pin and the placement, so the manuscript read exactly the text printed as Part I, and its citations of that text (Theorems 2.1 and 2.3, Research question 7) carry the right numbers. Its code, data and source notes are shipped with the prefix `02-boundary-phases-`; its $\LaTeX$ source, README, PDF and checksum list are not shipped and survive in the archive of the arrival commit. The manuscript is unrefereed and AI-assisted; its proofs are ordinary mathematical proofs, nothing in it is formalized in Lean or Rocq, and its place in this report confers no formal status.

What it answers. Theorem 14.1 answers the first half of Part I’s Research question 7 (D-finiteness at every fixed height), not its second half (Remark 14.3). Theorem 14.2 is a sharp result about a sub-sector of the boundary event of Theorem 4.1; it does not answer Research question 4, which asks for the asymptotic of the whole boundary error (Remark 14.4; the write’s Proposition 18.3 gives its leading term). Dated notes at both questions, at Remark 8.1 and at the end of Section 4.1 of Part I say so.

Overlap with Part I, credited. The two texts share 1.98 % of their word 8-grams (214 of 10,822). The state space (15.1) is (3.1), re-derived; the shifted hook product (18.2) is imported (Appendix A proves it); (19.9) is (5.14) with r variables; (20.7) extends (6.1) to a general success probability and is then mixed over the beta law; (20.11) are the Gaussian limits of (6.6); the Stirling factor (20.3) is the analogue of (5.16); Lemma 19.1 is a random-cut variant of Theorem 4.1, and Theorem 21.3 the analogue, for the rare event, of Theorem 9.1. The inversion of Section 22 uses the apparatus of Section 10 with a different rate and power, and Corollary 21.2 imports Theorem 2.1 and Corollary 2.2 ($b_{5,1} = 48/5$). Everything else is new to the repository. “The preceding (ProveIt, repository, asymptotic) report” in Part II means Part I.

Editorial changes. Section k of the manuscript is Section $k + 13$ here, its Appendices A and B are Appendices C and D, and its Research questions 1–8 are Research questions 8–15 (one counter runs through both Parts); theorem and equation numbers follow their sections. Every label carries the prefix `shr:bp:`, and one label was added (Remark 16.2). The manuscript’s $d = \binom{m-2}{2}$ is renamed d_R throughout, because Part I’s d is $\binom{m}{2}$; no normalization changes. Its three citations of its predecessor (the bibliography entry `prior`, whose pin and path this note records) now point to Part I. Its title, subtitle, result box, abstract and status paragraph, from its title page, open this Part. Falling factorials are typeset as in Part I ($n^{\bar{k}}$, the manuscript’s $(n)_k$); rising factorials keep the manuscript’s $(n)^{\bar{k}}$. The bibliographies are merged: the OEIS, Kauers–Koutschan and Sun entries are Part I’s (the manuscript inspected the OEIS entries on 4 October 2026 and arXiv v2 of Kauers–Koutschan, its Section 6.4 and Conjectures 18–19), and Bostan–Lairez–Salvy [8] is added. The [write] notes, Remarks 14.3 and 14.4, Section 18.3 with Proposition 18.3, and Section 24.1 are new. No other statement, proof or number of the manuscript was changed.

[write] Added 5 October 2026, batch 98: notation of Part II. Part II keeps the manuscript’s symbols except d_R , and several of them clash with Part I’s. Within Part II every symbol has the meaning below; the right column gives Part I’s meaning, the tempting false reading. Shared without change: $T_m(n)$, $F_m(z)$, $M_m(n)$, α_m , Δ , the hook count g^λ , $J_k = \prod_{j < k} j!$ (Part II uses it at $k = r$), W_{-1} , and $\mathbb{E}$.

symbol	meaning in Part II	not: meaning in Part I
d_R	$\binom{r}{2} = \binom{m-2}{2}$, the manuscript’s d	$d = \binom{m}{2}$
d_s, b_s, r_s	completions, births and active rows among the first s event letters (Section 16)	—
r	the size of an active block in Sections 15–17; $m - 2$ from Section 18 on	the degree of h_r , the index of p_r
$R_m(n)$	the rare count $\#\{D_1 < B_m\}$ (14.3)	$R_n(x)$ the pair weight; $R_m = m^{-m}$; $R(z)$
$H_m(n); H(a)$	the core count (18.1); the indicator $\binom{a}{a}$ of $a \geq 0$ (16.4)	$H(\lambda)$, Hermite H_j , H_{20}
$\mathcal{K}_m; K_r(u, v); K$	the rare constant (14.6); the chamber kernel; $\sum_i k_i$ in (20.1)	K_m (2.2); Krawtchouk K_j
$b_{m,j}^R; b_{5,1}^{\text{total}}$	rare coefficients; Part I’s $b_{5,1} = 48/5$	$b_{m,j}$ of (2.5)
$B_i, D_i; B, D; \mathbf{B}, \mathbf{D}$	birth and completion ranks $a_{i,1}, a_{i,n}$; event letters; the birth and completion maps	$B = m(m-1)^2(m-2)/12$; Bernoulli B_{2r} (also in (20.3))
$C_r(n)$	the strict chamber (15.2)	C_n the central event
$\mathcal{I}; X_i$	$\{1 \leq X_i \leq n-2\}$; $\text{Bin}(n, U)$ counts mixed over $U \sim \text{Beta}(n, n+1)$ (Section 19)	$\{1 \leq X_i \leq n-1\}$; independent $\text{Bin}(n, p)$
$W_n(x)$	the rare weight (19.3)	$R_n(x)$ plays this role
$Y_i; Z_{i,n}, V_n, V$	$(2X_i - n)/\sqrt{n}$ for the mixed X_i ; the components of (19.5)	Y_i the same formula for independent X_i ; Z_i standardized counts (Section 9)
$\varepsilon; \mathcal{E}_\varepsilon$	$n^{-1/2}$; the formal moment functional (20.2)	ε_j random signs; $\mathcal{E}_z$ (5.8)
$\Psi_r; Q_r(z); Q_m(n)$	the interaction factor (19.8); its normalized expectation (written $Q_r(0) = 1$ once in (20.2)); the common-phase count $T_{m,B^m D^m}(n)$ (21.2)	—
$A_m(z); \mathcal{A}_{5,L}, \mathcal{L}_5$	the Stirling factor (20.3); truncations and leading term of (21.1)	$A = \binom{m}{3}$, the event $\mathcal{A}$; $A_{5,L}, L_5$ for T_5

symbol	meaning in Part II	not: meaning in Part I
$S_1, S_2; S_h(t); S$	$\sum_i y_i$ and $\sum_i y_i^2$ (Section 20); bounded-height Dyck series; the event that row m starts after row 1 ends	$S_1 = h_1(q) = \sum_{i < j} (Y_i + Y_j)^2$
$p_i, q_i; p$ $P_j; P_1(y)$	rank marks (17.5); a fixed success probability in (20.7) monic orthogonal polynomials of $\text{Bin}(n, p)$ in x ; the first coefficient of Ψ_r (20.10)	p the threshold, p_r power sums, q_{ij} monic orthogonal $P_j(y)$ of the binomial Vandermonde ensemble (Section 6.2)
λ_m, β_m, a	$m \log m - \log 4, (m-2)^2/2, \log(y/K_m)$ (22.1)	$\lambda = m \log m, \beta_j$ recurrence coefficients, $L = \log(y/K)$
L	a truncation order; $ v - u $ in Lemma 15.1	a truncation order; $\log(y/K)$
$h(w)$	the maximal Dyck height of a word	$h_r(q), h_j$
$C_a(n, p), c_k, N_k$	centred binomial moments and series coefficients (Appendix D)	$c_{m,j}, C(z), C_{n-1}, C_{m,L}$
$\mathfrak{M}, \mathfrak{S}_r, \text{Diag}, \ell_s$	signed multinomial (15.3), the symmetric group, the diagonal, event labels (17.4)	—

14 The question and the scope of the answer

14.1 The OEIS family

For $m, n \geq 1$, let $T_m(n)$ be the number of arrays $A = (a_{i,j})_{1 \leq i \leq m, 1 \leq j \leq n}$ containing each of $1, \dots, mn$ once, with increasing rows, columns, diagonals, and downward antidiagonals. Equivalently, the following comparisons suffice whenever the cells exist:

$$a_{i,j} < a_{i,j+1}, \quad a_{i,j} < a_{i+1,j}, \quad a_{i,j+1} < a_{i+1,j}. \quad (14.1)$$

The diagonal comparison follows from row and column increase. We set $T_m(0) = 1$. This is the family A181196; the height-three, height-four, and height-five rows are A181197, A181198, and A181199 [1, 2, 3, 4]. The direction of the limit matters: m is fixed and n grows. A fixed-width strip theorem with a growing number of rows is a different statement.

Kauers and Koutschan discuss the height-four and height-five problems in Section 6.4 of their 2023 paper [7]. The OEIS entries inspected for this report still label their displayed recurrences as conjectural: order two and degree nine for A181198, and order three and degree twenty-four for A181199. The same paper gives related finite-sum conjectures. These specific formulas are not used anywhere in our proofs.

14.2 What the preceding repository report already established

The report *Fixed-height shifted rectangles* in ProveIt (Part I of the present report) obtains an all-order expansion for $T_m(n)$, including

$$T_m(n) \sim K_m m^{mn} n^{-\alpha_m}, \quad \alpha_m = \frac{m^2 - 1}{2}, \quad K_m = \frac{\sqrt{m} \prod_{j=0}^{m-1} j!}{4^{\binom{m}{2}} (2\pi)^{(m-1)/2}}. \quad (14.2)$$

It proves the leading asymptotic conjecture for A181199 without assuming its recurrence, and proves nonalgebraicity at every height $m \geq 3$. Its Question 7 asks whether the generating function is D-finite for every fixed height. Its coarse exponentially small boundary bound is explicitly not claimed sharp. These are the two gaps addressed here.

The repository snapshot used for the comparison is

db20eb37982abc0f163c6d308d392f5ac4c9bc62.

The source/status file in the package records the exact report path, inspected literature, and limited identifier searches. We do not infer historical novelty merely from an OEIS conjecture label or a repository question.

14.3 New statements developed in this manuscript

Our exact-enumeration result is stronger than existence of an unspecified recurrence.

Theorem 14.1 (Rational diagonality at every fixed height). *For every fixed integer $m \geq 1$,*

$$F_m(z) = \sum_{n \geq 0} T_m(n) z^n$$

is the diagonal of a rational power series over $\mathbb{Q}$. In particular it is D -finite, and $(T_m(n))_{n \geq 0}$ is P -recursive. More generally, each fixed ordering of row births and row completions has a rational-diagonal generating function, after assigning arbitrary rational values at $n = 0, 1$.

The proof, given in Sections 15–16, is a finite construction from chamber kernels. It provides at most $m(m-1)$ free integer summation indices for each event ordering. This count concerns summation variables, not the number of variables in a minimal rational function or the order of a minimal differential equation.

For $m \geq 2$, write

$$B_i = a_{i,1}, \quad D_i = a_{i,n}, \quad R_m(n) = \#\{A : D_1 < B_m\}, \quad n \geq 1, \quad (14.3)$$

and set $R_m(0) = 0$. Thus $R_m(n)$ counts arrays for which the first row finishes before the last row starts. When $n \geq 2$, its complement has all rows active simultaneously at some intermediate stage. Put

$$r = m - 2, \quad d_R = \binom{r}{2}, \quad J_r = \prod_{j=0}^{r-1} j!, \quad J_0 = 1. \quad (14.4)$$

Empty products are one, throughout. In the chamber formulas of Sections 15–17, r may instead denote a general active-block size; from Section 18 onward it is $m - 2$.

Theorem 14.2 (Sharp rare-boundary expansion). *For each fixed $m \geq 2$ and each fixed $L \geq 0$,*

$$R_m(n) = \mathcal{K}_m \left(\frac{m^m}{4} \right)^n n^{-r^2/2} \left(\sum_{j=0}^L \frac{b_{m,j}^R}{n^j} + O_{m,L}(n^{-L-1}) \right), \quad (14.5)$$

where

$$\mathcal{K}_m = \sqrt{\frac{m}{2}} (2\pi)^{-r/2} 4^{-d_R} 9^{-r} J_r, \quad b_{m,0}^R = 1, \quad (14.6)$$

and all $b_{m,j}^R$ are effectively computable rational numbers. The first correction is

$$b_{m,1}^R = \frac{r(6r^3 - 52r^2 - 268r - 283)}{72(r+2)}. \quad (14.7)$$

At $m = 2$, the assertion reduces to the exact identity $R_2(n) = 1$.

The proof occupies Sections 18–20. It is independent of (14.2). The latter is used only when we subsequently divide $R_m(n)$ by $T_m(n)$.

What has and has not been resolved. The fixed-height D-finiteness question is answered affirmatively by an exact formula, and a sharp rare-event sector is obtained at every height. The published candidate low-order recurrences and finite sums remain unproved in this manuscript. Our all-order series are Poincaré expansions with fixed height and fixed truncation order. No Borel summability, Stokes constants, or complete exponentially improved transseries is claimed.

Remark 14.3 ([write] Research question 7 of Part I is answered in part). Added 5 October 2026, batch 98. The abstract printed at the head of this Part says that Theorem 14.1 resolves “an explicit question in the preceding ProveIt report”, and the delivered README says “This resolves Question 7”. That overstates it. Research question 7 of Part I asks two things: whether F_m is D-finite for every fixed m , and “how does the minimal annihilating order grow with m ?”. Theorem 14.1 answers the first affirmatively. It says nothing about the second: the construction gives at most $m(m-1)$ summation indices, which the text above states is not a bound on the order of a minimal differential equation, Remark 16.2 says that no minimal operator is produced, and the manuscript’s own fourth research question (here Research question 11) asks for exactly this growth. The result box above (“The fixed-height D-finiteness question is answered affirmatively”) is accurate. The second half of Research question 7 stays open; a dated note at that question says so.

Remark 14.4 ([write] Research question 4 of Part I is not settled by R_m). Added 5 October 2026, batch 98. Section 14 lists the non-sharp boundary bound of Part I among “the two gaps addressed here”. The sharp question there, Research question 4, asks for the asymptotic of the boundary error $E_{m,n}(1/2)$ of Theorem 4.1. Part II does not determine it. Its rare count is a different and much smaller sector. In the order-statistics model of Section 4.1, the rare arrays have every row-1 sample below every row- m sample, so at the cut $p = 1/2$ either $X_1 = n$ or $X_m = 0$; hence the rare event lies inside the boundary event and

$$R_m(n) \leq M_m(n) E_{m,n}(1/2)$$

(Proposition 18.3(c), which also gives the growth of the ratio). Exact values, computed at intake from the definitions (row-state dynamic program for $T_m(n)$ and $R_m(n)$, the rational sum (4.5) for $I_m(n)$, and $M_m(n)E_{m,n}(1/2) = T_m(n) - M_m(n)I_m(n)$):

m	n	$M_m(n)E_{m,n}(1/2)/R_m(n)$	$2^n E_{m,n}(1/2)$	$E_{m,n}(1/2)/(2^{1-n}3^{1-m}J_{m-1}(4n)^{-\binom{m-1}{2}})$
3	8	4.892	$7.165 \cdot 10^{-3}$	1.0317
3	12	37.20	$4.835 \cdot 10^{-3}$	1.0444
3	16	357.5	$3.594 \cdot 10^{-3}$	1.0351
3	20	3907	$2.855 \cdot 10^{-3}$	1.0276
3	30	$2.053 \cdot 10^6$	$1.885 \cdot 10^{-3}$	1.0178
3	40	$1.328 \cdot 10^9$	$1.407 \cdot 10^{-3}$	1.0131
2	20	$8.962 \cdot 10^4$	0.6817	1.0226
2	80	$5.104 \cdot 10^{22}$	0.6704	1.0056
4	12	7.183	$1.381 \cdot 10^{-6}$	1.0311
4	20	374.2	$2.995 \cdot 10^{-7}$	1.0350

At $m = 3$ the boundary error is 4.9, 37, 357 and 3907 times the rare count at $n = 8, 12, 16, 20$, and the factor keeps growing roughly like 2^n . As a probability the boundary error is 2^{-n} times a power of n (second column), while $R_m(n)/M_m(n) \leq \binom{2n}{n}^{-1}$ is at most of order 4^{-n} by (18.5). So Theorem 14.2 is a sharp theorem about a sub-sector of the boundary event, not the sharp boundary asymptotic; Research question 4 stays open, re-scoped in a dated note there. The write’s Proposition 18.3 supplies its leading term (last column). The program used for the table is the intake’s and is not shipped; the same numbers follow from the shipped enumerator count of code/02-boundary-phases-tableaux.py and the sum (4.5).

15 Boundary events and the exact walk model

15.1 Prefix states, including both exceptional boundaries

Insert the labels $1, 2, \dots, mn$ in order, and let x_i be the number of filled cells in row i . Row increase makes the filled cells in each row a prefix. The state space is exactly

$$n \geq x_1 \geq x_2 \geq \dots \geq x_m \geq 0, \quad x_i = x_{i+1} \implies x_i \in \{0, n\}. \quad (15.1)$$

Indeed, if a lower row has $j < n$ filled cells, its last filled cell forces cell $j + 1$ in the row above to be filled. If the lower row is full, column increase forces the upper row to be full. Conversely these conditions make the filled set an order ideal of (14.1). Consequently paths from $(0, \dots, 0)$ to $(n, \dots, n)$ through (15.1), using unit steps $+e_i$, are in bijection with the arrays.

A row is *unborn*, *active*, or *completed* according as its coordinate is 0, belongs to $\{1, \dots, n - 1\}$, or is n . Completed rows form a prefix and unborn rows a suffix. Thus the active rows always form one consecutive block, and their coordinates are strictly decreasing. The ties at zero and at n cannot be discarded: forbidding either kind changes the enumeration.

For $n \geq 2$, birth and completion are different unit steps. Births occur in row order $1, \dots, m$, and completions also occur in that order. Record a letter B for a birth and D for a completion. Every array gives a word w of length $2m$ with m letters of each kind and with at least as many B 's as D 's in every prefix. Such a word is a Dyck word, with B interpreted as an up-step. Conversely the i th B means the birth of row i , and the i th D means the completion of row i ; no further row-labeling data are needed.

15.2 The strict chamber between two events

For $r \geq 0$ define

$$C_r(n) = \{(u_1, \dots, u_r) : n - 1 \geq u_1 > \dots > u_r \geq 1\}, \quad C_0(n) = \{()\}. \quad (15.2)$$

During an interval with no birth or completion, its r active coordinates move by $+e_i$ in this chamber. If $u, v \in C_r(n)$, write $K_r(u, v)$ for the number of such walks from u to v , including a zero-length walk when $u = v$.

For an integer vector $a = (a_1, \dots, a_r)$ define the everywhere-defined integer multinomial

$$\mathfrak{M}(a) = \begin{cases} (a_1 + \dots + a_r)! / (a_1! \dots a_r!), & a_i \geq 0 \text{ for all } i, \\ 0, & \text{otherwise,} \end{cases} \quad \mathfrak{M}() = 1. \quad (15.3)$$

Lemma 15.1 (Reflection kernel). *For strict endpoints $u, v \in C_r(n)$,*

$$K_r(u, v) = \sum_{\sigma \in \mathfrak{S}_r} \text{sgn}(\sigma) \mathfrak{M}(v_1 - u_{\sigma(1)}, \dots, v_r - u_{\sigma(r)}). \quad (15.4)$$

Equivalently, when $L = |v| - |u| \geq 0$,

$$K_r(u, v) = L! \det_{1 \leq i, j \leq r} \frac{1}{(v_i - u_j)!}, \quad (15.5)$$

where $1/k! = 0$ for negative integers k . For $r = 0$, the value is one.

Proof. Consider the signed family of all unrestricted $+e_i$ walks from a permutation of u to v . The number starting at $(u_{\sigma(1)}, \dots, u_{\sigma(r)})$ is the multinomial in (15.4). Cancel every walk that has a collision of two coordinates: choose its earliest collision time, breaking any tie by a fixed ordering of coordinate pairs, and exchange the histories of the colliding coordinates up to that time. The starting permutation is changed by a transposition, the endpoint and step set are unchanged, and the operation is an involution. Before the chosen time, all equality relations are preserved under this exchange, so the same collision is selected on the second application.

A walk remaining after cancellation has no collisions. Its coordinate order cannot change, since a unit step cannot cross an integer coordinate without meeting it. Its strictly decreasing endpoint forces its starting permutation to be the identity, and it stays in the decreasing chamber. All coordinates are between 1 and $n - 1$: along an uncanceled walk each coordinate is nondecreasing and lies between its own endpoints. Thus no extra wall reflection is needed for the finite upper bound. The surviving walks are exactly those counted by $K_r(u, v)$. $\square$

The proof is the classical reflection argument, included to specify precisely which walls are present. Formula (15.4), rather than an unqualified factorial quotient, is the appropriate form for subsequent binomial-sum manipulations: it is defined even when an individual displacement is negative.

15.3 Birth and completion maps

If $v \in C_r(n)$, a birth changes the active vector by

$$B(v) = (v_1, \dots, v_r, 1), \quad \text{allowed if } r = 0 \text{ or } v_r \geq 2. \quad (15.6)$$

A completion changes it by

$$D(v) = (v_2, \dots, v_r), \quad \text{allowed if } r \geq 1 \text{ and } v_1 = n - 1. \quad (15.7)$$

The completion step increases the old first active coordinate from $n - 1$ to n before removing it. All previously completed coordinates are already n . These two maps account exactly for the exceptional boundary ties.

16 Finite binomial sums and rational diagonals

16.1 The event-skeleton formula

Fix a Dyck word $w = w_1 \cdots w_{2m}$. Let b_s, d_s be the numbers of births and completions in its first s letters, and set $r_s = b_s - d_s$, for $0 \leq s \leq 2m$. Let u_s be the active vector immediately after event s , and v_s the active vector immediately before event $s + 1$. The initial vector is $u_0 = ()$. A skeleton consists of vectors $v_s \in C_{r_s}(n)$ with

$$u_{s+1} = \begin{cases} B(v_s), & w_{s+1} = B, \\ D(v_s), & w_{s+1} = D, \end{cases} \quad 0 \leq s < 2m, \quad (16.1)$$

subject to the corresponding legality conditions. All u_s are thereby affine expressions in earlier endpoint coordinates and the constants $1, n$. Define $T_{m,w}(n)$ to count arrays with this event word.

Theorem 16.1 (Exact finite-phase enumeration). *For $n \geq 2$,*

$$T_{m,w}(n) = \sum_{\text{legal skeletons for } w} \prod_{s=0}^{2m-1} K_{r_s}(u_s, v_s), \quad T_m(n) = \sum_{w \in \mathcal{D}_m} T_{m,w}(n), \quad (16.2)$$

where $\mathcal{D}_m$ denotes the Dyck words with m births. Each first sum can be written with at most $m(m-1)$ free integer indices, each ranging between 0 and n , and with only affine equality and inequality constraints.

Proof. Cut a valid prefix path immediately before and after every boundary event. The intervening pieces are chamber walks with the stated endpoints. Conversely, choosing a legal skeleton and one chamber walk in each interval reconstructs a unique full prefix path. The events each contribute a single unit step, and the intervals concatenate in a prescribed order; there is no additional interleaving factor. This proves the product and sum.

There are initially $\sum_{s=0}^{2m-1} r_s$ endpoint slots. This sum is at most m^2 : at each position s the height is at most $\min(s, 2m-s)$, with equality throughout for $B^m D^m$. Each of the m completion events fixes one different endpoint coordinate to $n-1$. Eliminating these coordinates leaves at most $m^2 - m$ free slots. The chamber bounds, strict gaps, birth conditions, and all endpoint substitutions are affine in n and the remaining coordinates. $\square$

As a consistency check, every skeleton with nonzero weight satisfies

$$\sum_{s=0}^{2m-1} (|v_s| - |u_s|) = m(n-2). \quad (16.3)$$

Each of the m rows has precisely $n-2$ non-event steps. Equivalently, telescoping the active sums adds one at a birth and removes $n-1$ at a completion, which gives the same identity.

16.2 An explicit binomial encoding of every constraint

For integers a, b , use $\binom{a}{b} = [t^b](1+t)^a$ when $b \geq 0$, and zero when $b < 0$. In particular,

$$H(a) = \binom{a}{a} = \mathbf{1}_{\{a \geq 0\}}, \quad \delta(a) = H(a)H(-a) = \mathbf{1}_{\{a=0\}}. \quad (16.4)$$

Every affine inequality or equality in (16.2) is therefore a product of such factors. For example, strictness $v_i > v_{i+1}$ is $H(v_i - v_{i+1} - 1)$, and a completion uses $\delta(v_1 - n + 1)$. Moreover, for $r \geq 1$,

$$\mathfrak{M}(a_1, \dots, a_r) = \left(\prod_{i=1}^r H(a_i) \right) \prod_{i=1}^{r-1} \binom{a_i + \dots + a_r}{a_i}. \quad (16.5)$$

If a displacement is negative the leading indicator makes the term zero; all other factors remain defined. If all displacements are nonnegative, the binomial product telescopes to the multinomial.

Expand the fixed-size permutation sums in (15.4), replace each multinomial by (16.5), sum each free coordinate from 0 to n , and insert the legality factors. The result is a finite linear combination of bounded multiple sums of products of binomial coefficients with affine arguments. There are no reciprocal binomial coefficients, no factorial poles, and no hidden variable number of summation indices.

16.3 The imported diagonal theorem and its application

We use the following result of Bostan, Lairez, and Salvy [8, Theorem 3.5 and Corollary 3.6]: a univariate multiple binomial sum has a generating function that is the diagonal of a rational power series; such sequences are P-recursive. Their class is closed under affine index substitutions, products, sums, and finite partial summation.

For clarity, consider a rational function regular at the origin, with expansion

$$Q(x_1, \dots, x_k) = \sum q_{n_1, \dots, n_k} x_1^{n_1} \cdots x_k^{n_k}.$$

Its diagonal is

$$\text{Diag } Q(z) = \sum_{n \geq 0} q_{n, \dots, n} z^n.$$

A power series is D-finite if it satisfies a nonzero linear differential equation with polynomial coefficients. Equivalently its coefficient sequence satisfies a linear recurrence with polynomial coefficients.

Proof of Theorem 14.1. The preceding encoding exhibits $T_{m,w}(n)$, for $n \geq 2$, as a multiple binomial sum. To define a sequence on all integers, multiply by $H(n-2)$ and assign zero below two. The sums from 0 to n can first be interpreted as the extended partial sums of the binomial-sum algebra; the multiplier removes their values outside the intended range. Adding any finitely supported correction preserves that algebra, since shifted Kronecker deltas are allowed. The imported theorem gives a rational diagonal for each event-word sequence. Summing the finitely many words and adding the terms at $n = 0, 1$ gives F_m .

This argument proves rational diagonality, not merely a heuristic closure claim about hypergeometric expressions. D-finiteness and P-recursiveness then follow from the same theorem. At $m = 1$, one may also observe directly that $F_1(z) = 1/(1-z)$. $\square$

Remark 16.2 (Constructive does not mean a compact telescope is supplied). The skeleton and binomial formulas are explicit finite input to the rational-integral and diagonal constructions of [8]. This is an effective existence proof. We have not carried out that elimination to produce a small rational function or a minimal differential operator for general m . In particular, it does not certify that either operator currently displayed in OEIS is the correct one.

Combined with the previous report's algebraicity classification (Part I, Theorem 2.3), the result places the entire family in a precise hierarchy: F_m is algebraic for $m \leq 2$, and transcendental but D-finite for $m \geq 3$. The nonalgebraicity part of that statement is inherited from the previous report, not reproved or claimed as new here.

17 Which event words occur, and which statistics remain D-finite?

17.1 A complete realizability criterion

Let $h(w)$ be the maximum Dyck height of w .

Theorem 17.1 (Event-word realizability). *For $m \geq 1$, $n \geq 2$, and $w \in \mathcal{D}_m$,*

$$T_{m,w}(n) > 0 \iff h(w) \leq n - 1. \quad (17.1)$$

Consequently every Dyck word of semilength m occurs as soon as $n \geq m + 1$.

Proof. At height r there must be r distinct active coordinates in $\{1, \dots, n-1\}$, so $r \leq n-1$ is necessary.

For sufficiency, follow the prescribed word. Before a birth at active height r , the height bound implies $r \leq n-2$. Move the existing coordinates, in top-to-bottom order, to $(n-1, n-2, \dots, n-r)$. Each target dominates the present coordinate, since any decreasing vector in $C_r(n)$ has $u_i \leq n-i$. The moves are legal because the coordinate above has already reached its target. The last target is at least two, so a new coordinate

one can be appended. Before a completion, simply move the top active coordinate to $n - 1$ and complete that row. Neither operation creates any additional boundary event. Continuing to the end constructs a valid path with exactly w . $\square$

The number of event words available at width n is thus a classical bounded-height Dyck number. If $S_h(t)$ is its generating function by semilength with height at most h , the first-return decomposition gives

$$S_0(t) = 1, \quad S_h(t) = \frac{1}{1 - tS_{h-1}(t)} \quad (h \geq 1). \quad (17.2)$$

This counts possible event types, not arrays: the sector weights are generally far from equal.

At $n = 2$ there is only the alternating word $(BD)^m$, and $T_m(2) = 1$. At $n = 3$, height is at most two. A one-row active phase has only one coordinate to move, while a two-row active phase has the unique vector $(2, 1)$. Thus each permitted word has weight one. Formula (17.2) gives $T_m(3) = 2^{m-1}$ for $m \geq 1$, recovering a known small-width formula [9], not a new one.

17.2 Small worked examples

For two rows and $n \geq 2$, the alternating word has weight one. For the other word,

$$\begin{aligned} T_{2,BBDD}(n) &= \sum_{a=2}^{n-1} \sum_{b=1}^{n-2} K_2((a, 1), (n-1, b)) \\ &= \sum_{a=2}^{n-1} \sum_{b=1}^{n-2} (\mathfrak{M}(n-1-a, b-1) - \mathfrak{M}(n-2, b-a)). \end{aligned} \quad (17.3)$$

This is an explicit binomial sum without exceptional denominator values. Adding one recovers $T_2(n) = C_{n-1}$, the usual Catalan control.

At $m = 3, n = 4$, the sector weights are

Event word	<i>BBBDDD</i>	<i>BBDBDD</i>	<i>BBDDBD</i>	<i>BDBBDD</i>	<i>BDBBDD</i>
Number of arrays	4	16	4	4	1

Their sum is 29, and the rare subcount is 25. This also illustrates that having a full-height phase is possible but not yet typical at small width.

17.3 Birth and completion ranks

In a skeleton, immediately before event $s + 1$ there are d_s completed rows and $|v_s|$ cells in active rows. Hence its label is the affine form

$$\ell_{s+1} = d_s n + |v_s| + 1. \quad (17.4)$$

Let $p_1, \dots, p_m, q_1, \dots, q_m$ be independent indeterminates and define

$$T_m(n; \mathbf{p}, \mathbf{q}) = \sum_A \prod_{i=1}^m p_i^{B_i} q_i^{D_i}. \quad (17.5)$$

The birth and completion ranks are labels, not column positions.

Corollary 17.2 (Marked enumeration and moment totals). *For fixed m , the generating function in n of $T_m(n; \mathbf{p}, \mathbf{q})$ is a rational diagonal over $\mathbb{Q}(\mathbf{p}, \mathbf{q})$. For every fixed polynomial P with rational coefficients in the $2m$ event ranks,*

$$n \mapsto \sum_A P(B_1, \dots, B_m, D_1, \dots, D_m)$$

is a multiple binomial sum and is P -recursive. The same statements hold within each fixed event sector.

Proof. Insert the weights from (17.4) into (16.2). They are geometric sequences with affine exponents over the indicated coefficient field. Polynomial weights are also in the binomial-sum algebra, since an integer index is $\binom{k}{1}$. The same closure proof applies. When $n = 1$, use the appropriate finitely supported marked term; birth and completion coincide within a row only in this exceptional case. $\square$

This proves the assertion for *unnormalized moment totals*. An expectation divides by $T_m(n)$; P -recursiveness of such quotients is not asserted. Rotation of the array by 180 degrees followed by replacement of each label a by $mn + 1 - a$ gives the useful exact symmetry

$$B_i \longleftrightarrow mn + 1 - D_{m+1-i}, \quad T_{m,w}(n) = T_{m,\overline{w}^{\text{rev}}}(n), \quad (17.6)$$

where the bar interchanges B and D .

18 A first-completion cut for the rare sector

18.1 The dominant rare configuration

For $m \geq 2$ and $n \geq 2$, define the core subcount

$$H_m(n) = \#\{A : B_{m-1} < D_1 < B_m\}. \quad (18.1)$$

For $m = 2$ this equals $R_2(n) = 1$. For $m \geq 3$, the remainder has $D_1 < B_{m-1}$: the first row finishes while at least the last two rows are still unborn. This extra separation will give an exponentially smaller rate.

The classical shifted hook product for a strict positive partition $\lambda = (\lambda_1 > \dots > \lambda_k > 0)$ is

$$g^\lambda = \frac{|\lambda|!}{\prod_i \lambda_i!} \prod_{i < j} \frac{\lambda_i - \lambda_j}{\lambda_i + \lambda_j}. \quad (18.2)$$

It counts chains of strict partitions obtained by adding one cell at a time, equivalently standard shifted tableaux; see [9]. This is an imported classical enumeration formula, not a new hook theorem.

Proposition 18.1 (Exact core sum). *With $r = m - 2$, for all $m, n \geq 2$,*

$$H_m(n) = \sum_{n-2 \geq x_1 > \dots > x_r \geq 1} g^{(n-1, x_1, \dots, x_r)} g^{(n, n-x_r, \dots, n-x_1)}. \quad (18.3)$$

For $r = 0$ the sum has one term, equal to one.

Proof. Immediately before the first completion, row 1 has length $n - 1$ and the last row is empty. In the core sector every intervening row is active, so their lengths are $n - 2 \geq x_1 > \dots > x_r \geq 1$. Before this point no row has completed. Removing zero rows, the prefix path is a chain of strict partitions ending at $(n - 1, x_1, \dots, x_r)$, so it has the first hook count in (18.3).

The next step completes row 1. Reverse all remaining steps and rotate rows; the complementary lengths are $(n, n - x_r, \dots, n - x_1)$, a strict positive partition. Every reversed prefix stays in the strict-partition region until the endpoint. In particular, a coordinate equal to n at the endpoint causes no repeated completed-row boundary, since all other parts are strictly smaller. Thus the suffix has the second hook count. Their label sets are fixed by the size of the prefix, and the intervening completion step is unique. The two chains concatenate bijectively to the core arrays. $\square$

18.2 Two useful nonasymptotic error bounds

Write

$$M_m(n) = \frac{(mn)!}{(n!)^m}. \quad (18.4)$$

This counts all interleavings of m individually sorted rows, before imposing cross-row comparisons.

Proposition 18.2 (Separation bounds). *For $m \geq 2$ and $n \geq 1$,*

$$R_m(n) \leq \frac{M_m(n)}{\binom{2n}{n}}. \quad (18.5)$$

For $m \geq 3$ and $n \geq 2$,

$$0 \leq R_m(n) - H_m(n) \leq \frac{M_m(n)}{\binom{3n}{n}}. \quad (18.6)$$

For $m = 2$ the difference is zero.

Proof. In a uniform row interleaving, restrict attention to rows 1 and m . Of their $\binom{2n}{n}$ possible relative words, exactly one places all of row 1 before all of row m . Dropping other comparisons bounds the rare sector and proves (18.5).

For the remainder, rows $m - 1$ and m must both start after row 1 has finished. Among the relative interleavings of these three rows, the probability that the first n letters all come from row 1 is

$$\frac{(2n)!/(n!)^2}{(3n)!/(n!)^3} = \frac{1}{\binom{3n}{n}}.$$

Again dropping all other constraints gives an upper bound. $\square$

These estimates have explicit constants and hold at finite n . Once the core asymptotic is known, (18.6) is smaller than the core by a polynomial factor times $(16/27)^n$. It is therefore negligible at every inverse-power order on the rare scale. We do not claim that (18.6) gives the sharp next exponential sector.

18.3 [write] The boundary error of Part I's central cut

[write] **Added 5 October 2026, batch 98.** This subsection is the write's, not the manuscript's. It gives the leading asymptotic of the boundary error asked for in Research question 4 of Part I, and places the rare sector of this Part inside that error (Remark 14.4). It uses only Part I's model and estimates and the hook argument of Proposition 18.1. Like the rest of the report it is unrefereed; the exact identity (a) and the asymptotic (b) were checked against exact values for $m = 2, 3, 4$ (table of Remark 14.4).

Proposition 18.3 ([write] Leading boundary error at the central cut). *Let $m \geq 1$, $n \geq 2$, and let $E_{m,n}(1/2)$ be the boundary error of Theorem 4.1 at $p = 1/2$. Let $X'_1, \dots, X'_{m-1}$ be independent $\text{Bin}(n, 1/2)$ variables, $\mathcal{I}' = \{1 \leq X'_i \leq n-1 \text{ for all } i\}$, and*

$$W'_n(x) = \prod_{1 \leq i < j \leq m-1} \frac{(x_i - x_j)^2}{(x_i + x_j)(2n - x_i - x_j)} \prod_{i=1}^{m-1} \frac{x_i}{2n - x_i}.$$

(a) Exactly,

$$E_{m,n}(1/2) = \frac{2^{1-n}}{(m-1)!} \mathbb{E}[\mathbf{1}_{\mathcal{I}'} W'_n(X')] + D_{m,n}, \quad 0 \leq D_{m,n} \leq 2m(m-1)4^{-n}. \quad (18.7)$$

(b) As $n \rightarrow \infty$,

$$E_{m,n}(1/2) \sim 2^{1-n} 3^{1-m} J_{m-1} (4n)^{-\binom{m-1}{2}}, \quad J_{m-1} = \prod_{j=0}^{m-2} j!. \quad (18.8)$$

(c) Consequently $M_m(n)E_{m,n}(1/2)/T_m(n) \sim 2^{1-n} 3^{1-m} (4n)^{m-1}/(m-1)!$, and, for $m \geq 2$, $R_m(n) \leq M_m(n)E_{m,n}(1/2)$ for every $n \geq 2$, with

$$\frac{M_m(n)E_{m,n}(1/2)}{R_m(n)} \sim \frac{2 \cdot 3^{m-3} (m-2)!}{\sqrt{\pi} 4^{m-2}} 2^n n^{-(2m-3)/2}.$$

Proof. Use the order-statistics model of Section 4.1 at $p = 1/2$: X_i counts the row- i samples below $1/2$, the X_i are independent $\text{Bin}(n, 1/2)$, and the proof of Theorem 4.1 identifies $E_{m,n}(1/2)$ with $\mathbb{P}(\mathcal{A} \cap \mathcal{I}^c)$.

(a) On $\mathcal{A}$ the cells with values below $1/2$ form an order ideal, so X is a prefix state (3.1): $X_i = 0$ forces $X_{i+1} = \dots = X_m = 0$, and $X_i = n$ forces $X_1 = \dots = X_i = n$. Hence on $\mathcal{A} \cap \mathcal{I}^c$ either exactly one coordinate lies in $\{0, n\}$, and then it is $X_m = 0$ or $X_1 = n$ with every other coordinate in $[1, n-1]$, or at least two coordinates lie in $\{0, n\}$. The second event, whose probability on $\mathcal{A}$ we call $D_{m,n}$, has probability at most $\binom{m}{2} (2 \cdot 2^{-n})^2 = 2m(m-1)4^{-n}$.

Let $x = (x_1, \dots, x_{m-1}, 0)$ with $n-1 \geq x_1 > \dots > x_{m-1} \geq 1$. As in Section 4.1, $\mathbb{P}(\mathcal{A} \mid X = x)$ is the product of a prefix and a suffix probability. Every prefix path to x has last coordinate 0 and first coordinate below n , so its states are strict partitions and the prefix count is $g^{(x_1, \dots, x_{m-1})}$. The rotated complement (3.3) of x is $(n, n-x_{m-1}, \dots, n-x_1)$, a strict positive partition in which only the first part can reach n ; as in the proof of Proposition 18.1, the suffix count is $g^{(n, n-x_{m-1}, \dots, n-x_1)}$. Dividing both counts by the multinomial numbers of interleavings and using (3.5), the pairs among $x_1, \dots, x_{m-1}$ contribute $\frac{x_i - x_j}{x_i + x_j} \cdot \frac{x_i - x_j}{2n - x_i - x_j}$ and the pairs of the part n with the parts $n - x_i$ contribute $\frac{x_i}{2n - x_i}$, so $\mathbb{P}(\mathcal{A} \mid X = x) = W'_n(x_1, \dots, x_{m-1})$. The weight W'_n is symmetric and vanishes at ties; summing over decreasing vectors and multiplying by $\mathbb{P}(X_m = 0) = 2^{-n}$, the stratum $\{X_m = 0\}$ contributes $2^{-n} \mathbb{E}[\mathbf{1}_{\mathcal{I}'} W'_n(X')]/(m-1)!$. Reflecting every sample $u \mapsto 1-u$ and reversing the rows preserves the model and $\mathcal{A}$ and maps X to $(n - X_m, \dots, n - X_1)$, so the stratum $\{X_1 = n\}$ contributes the same. This proves (18.7). (For $m = 1$ there are no pairs, $W'_n = 1$, $D_{1,n} = 0$, and $E_{1,n}(1/2) = 2^{1-n}$.)

(b) Put $c = \binom{m-1}{2}$, $Y_i = (2X'_i - n)/\sqrt{n}$ and $C'_n = \{|Y_i| \leq \sqrt{n}/2 \text{ for all } i\}$; then $C'_n \subset \mathcal{I}'$. On C'_n , exactly as in (5.2),

$$(4n)^c W'_n(X') = \Delta(Y)^2 \prod_{i < j} \left(1 - \frac{(Y_i + Y_j)^2}{4n}\right)^{-1} \prod_i \frac{1 + Y_i/\sqrt{n}}{3 - Y_i/\sqrt{n}},$$

where every pair factor lies in $[1, 4/3]$ and every single factor in $[1/7, 3/5]$. The right side is at most $(4/3)^c \Delta(Y)^2$, a uniformly integrable family because the moments of each fixed degree of the Y_i are bounded uniformly in n (Section 5.4), and it converges to $3^{1-m} \Delta(y)^2$ uniformly on compact sets. Since Y converges

in distribution to a vector G of independent standard normals, $(4n)^c \mathbb{E}[\mathbf{1}_{C'_n} W'_n(X')] \rightarrow 3^{1-m} \mathbb{E} \Delta(G)^2 = 3^{1-m} (m-1)! J_{m-1}$ by (5.14) in $m-1$ variables. Off C'_n , $0 \leq W'_n \leq 1$ (Remark 4.2 and $x_i \leq n$), and $\mathbb{P}(C'_n)^c \leq 2(m-1)e^{-n/8}$ by (5.10), which is $o(n^{-c})$. Inserting this into (18.7), where $D_{m,n} = O(4^{-n})$ is $o(2^{-n}n^{-c})$, gives (18.8).

(c) The first relation is (18.8) divided by $T_m(n)/M_m(n) \sim J_m(4n)^{-\binom{m}{2}}$ (2.3), since $J_m = (m-1)! J_{m-1}$ and $\binom{m}{2} - \binom{m-1}{2} = m-1$. In the same model the rare event $\{D_1 < B_m\}$ is the event that every row-1 sample is below every row- m sample. If the largest row-1 sample is below $1/2$, then $X_1 = n$; otherwise the smallest row- m sample exceeds $1/2$ and $X_m = 0$. Either way $X \notin \mathcal{I}$, so $R_m(n)/M_m(n) = \mathbb{P}(\mathcal{A} \cap \{D_1 < B_m\}) \leq \mathbb{P}(\mathcal{A} \cap \mathcal{I}^c) = E_{m,n}(1/2)$. The ratio follows from (18.8), Theorem 14.2 and $M_m(n) \sim \sqrt{m}(2\pi n)^{(1-m)/2} m^{mn}$ (Section 5.5): the powers of m^m cancel, $\sqrt{m}/\sqrt{m/2} = \sqrt{2}$, $(2\pi)^{(1-m)/2 + (m-2)/2} = (2\pi)^{-1/2}$, $3^{1-m} 9^{m-2} = 3^{m-3}$, $J_{m-1}/J_{m-2} = (m-2)!$, $4^{-\binom{m-1}{2} + \binom{m-2}{2}} = 4^{2-m}$, and the power of n is $-\binom{m-1}{2} + \frac{1-m}{2} + \frac{(m-2)^2}{2} = -\frac{2m-3}{2}$. $\square$

At $m=2$, (c) reads $\binom{2n}{n} E_{2,n}(1/2) \sim \frac{2}{3\sqrt{\pi}} 2^n n^{-1/2}$, since $R_2(n) = 1$. Part (a) is also the first step of an all-order expansion: the expectation in (18.7) is an interior sum of Part I's type with $m-1$ rows and one extra rational factor per row, so the method of Section 5 should expand it to every order in $1/n$. That expansion is not carried out here (Section 24.1).

19 A beta-binomial representation and the leading constant

19.1 A random cut at the first completion

Take m independent rows of n independent uniform random variables on $(0, 1)$ and sort within each row. Their global ranks are uniform among the $M_m(n)$ row interleavings. Let U be the maximum of the top row, and let $\mathcal{S}$ be the event that every entry in the bottom row exceeds U . Its probability is $\binom{2n}{n}^{-1}$. The joint density of U and $\mathcal{S}$ is $nu^{n-1}(1-u)^n$. Consequently

$$U \mid \mathcal{S} \sim \text{Beta}(n, n+1). \quad (19.1)$$

Conditional on $U = u$ and $\mathcal{S}$, the numbers of middle-row samples below u are independent variables

$$X_i \mid U = u \sim \text{Bin}(n, u), \quad 1 \leq i \leq r. \quad (19.2)$$

The X_i are exchangeable but *not* independent after integrating out U .

Set $\mathcal{I} = \{1 \leq X_i \leq n-2 \text{ for all } i\}$ and define

$$W_n(x) = \prod_{i < j} \frac{(x_i - x_j)^2}{(x_i + x_j)(2n - x_i - x_j)} \prod_{i=1}^r \frac{(n-1-x_i)x_i}{(n-1+x_i)(2n-x_i)}. \quad (19.3)$$

Denominators are positive on $\mathcal{I}$; equal coordinates give zero. Every pair factor and every single-coordinate factor belongs to $[0, 1]$, so $0 \leq W_n \leq 1$ there.

Lemma 19.1 (Exact mixture identity). *For all $m, n \geq 2$,*

$$\frac{H_m(n)}{M_m(n)} = \frac{1}{r! \binom{2n}{n}} \mathbb{E}[\mathbf{1}_{\mathcal{I}} W_n(X)], \quad (19.4)$$

with the beta-binomial law (19.1)–(19.2).

Proof. Given the counts x , the samples below U in the top and middle rows are independent sorted uniforms. Relative to their unconstrained multinomial interleavings, the first hook count contributes

$$\prod_i \frac{n-1-x_i}{n-1+x_i} \prod_{i < j} \frac{x_i - x_j}{x_i + x_j}$$

when $x_1 > \dots > x_r$. Above U , the complementary hook count similarly contributes

$$\prod_i \frac{x_i}{2n-x_i} \prod_{i < j} \frac{x_i - x_j}{2n-x_i-x_j}.$$

All cross-cut comparisons are automatic for the legal prefix state. The product is $W_n(x)$. Distinct middle counts have exactly one decreasing ordering; the mixture law and the weight are symmetric in those counts. Thus summing over decreasing counts is $1/r!$ times the unrestricted expectation on $\mathcal{I}$. Multiplying by $\mathbb{P}(\mathcal{S})$ proves the identity. This is also a probabilistic derivation of (18.3). $\square$

19.2 Scaling and the common Gaussian mode

Let

$$\varepsilon = n^{-1/2}, \quad Y_i = \frac{2X_i - n}{\sqrt{n}}, \quad V_n = 2\sqrt{n}(U - \tfrac{1}{2}), \quad Z_{i,n} = \frac{2(X_i - nU)}{\sqrt{n}}. \quad (19.5)$$

Then $Y_i = Z_{i,n} + V_n$. A direct Laplace expansion of the beta density gives $V_n \Rightarrow V \sim N(0, 1/2)$. Conditional binomial characteristic functions, uniformly for U near $1/2$, give joint convergence

$$(Z_{1,n}, \dots, Z_{r,n}, V_n) \Rightarrow (Z_1, \dots, Z_r, V), \quad (19.6)$$

where the Z_i are independent $N(0, 1)$ variables and are independent of V . For example, expanding $(1 - U + Ue^{2it/\sqrt{n}})^n e^{-2itnU/\sqrt{n}}$ gives the limiting factor $e^{-t^2/2}$ on $U = 1/2 + o(1)$; the beta mass outside that neighborhood tends to zero. The same argument applies to a fixed vector of t_i .

[write] Added 5 October 2026, batch 98: the convergence (19.6) in detail. The two sentences above are a sketch; here is the argument with its uniformity made explicit. The density of U given $\mathcal{S}$ is $f(u) = n \binom{2n}{n} u^{n-1} (1-u)^n$, so for $|v| < \sqrt{n}$ the density of V_n is

$$\sqrt{n} \binom{2n}{n} 4^{-n} \left(1 - \frac{v}{\sqrt{n}}\right) \left(1 - \frac{v^2}{n}\right)^{n-1} \rightarrow \frac{e^{-v^2}}{\sqrt{\pi}},$$

the $N(0, 1/2)$ density, and Scheffé's lemma gives $V_n \Rightarrow V$. Conditionally on $U = u$ the $Z_{i,n}$ are independent with characteristic function $\varphi_n(t; u) = (\mathbb{E} e^{ih(\xi - u)})^n$, where $h = 2t/\sqrt{n}$ and ξ is Bernoulli(u). The cumulant function $\kappa(h) = \log \mathbb{E} e^{ih(\xi - u)}$ has $\kappa(0) = \kappa'(0) = 0$, $\kappa''(0) = -u(1-u)$ and, since $|\mathbb{E} e^{ih(\xi - u)}| \geq 1 - h^2/8$, a third derivative bounded uniformly in $u \in [0, 1]$ for $|h| \leq 1/2$. Hence $\log \varphi_n(t; u) = -2u(1-u)t^2 + O(|t|^3/\sqrt{n})$ uniformly in u , for $|t| \leq \sqrt{n}/4$. As $|\varphi_n| \leq 1$,

$$\mathbb{E} \exp\left(isV_n + i \sum_i t_i Z_{i,n}\right) = \mathbb{E} \left[e^{isV_n} e^{-2U(1-U) \sum_i t_i^2} \right] + o(1) \rightarrow \mathbb{E} e^{isV} e^{-\sum_i t_i^2/2},$$

because $2U(1-U) = \frac{1}{2} - V_n^2/(2n)$ and $V_n \Rightarrow V$. This is (19.6).

Let $\Delta(y) = \prod_{i < j} (y_i - y_j)$. Algebraic substitution into (19.3) gives the exact factorization

$$W_n(X) = (4n)^{-d_{\mathbb{R}} 9^{-r}} \Delta(Y)^2 \Psi_r(\varepsilon, Y) \quad \text{on } \mathcal{I}, \quad (19.7)$$

where

$$\begin{aligned} \Psi_r(\varepsilon, y) &= \prod_{i < j} \left(1 - \frac{\varepsilon^2 (y_i + y_j)^2}{4} \right)^{-1} \\ &\quad \times \prod_{i=1}^r \frac{9(1 - 2\varepsilon^2 - \varepsilon y_i)(1 + \varepsilon y_i)}{(3 - 2\varepsilon^2 + \varepsilon y_i)(3 - \varepsilon y_i)}. \end{aligned} \quad (19.8)$$

The factor 9^{-r} has a simple origin: at $x_i \sim n/2$, the interaction with the completed top row and the unborn bottom row tends to $(1/3)(1/3) = 1/9$ for each middle row.

19.3 Gaussian normalization

For independent standard normal Z_i ,

$$\mathbb{E} \Delta(Z)^2 = r! J_r. \quad (19.9)$$

One elementary proof writes the Vandermonde determinant in the basis of monic Hermite polynomials. Their squared norms for the standard normal measure are $j!$; determinant expansion and orthogonality give $r! \prod_{j=0}^{r-1} j!$. The Hermite norms themselves follow from

$$\mathbb{E} \exp(tZ - t^2/2) \exp(sZ - s^2/2) = e^{ts}.$$

Since adding a common scalar does not change a Vandermonde,

$$\Delta(Z_1 + V, \dots, Z_r + V) = \Delta(Z). \quad (19.10)$$

Thus the extra beta fluctuation contributes a common Gaussian shift, but does not alter the leading Vandermonde normalization.

For completeness, moment convergence is justified by uniform tails. The beta density is proportional to $[u(1-u)]^{n-1}(1-u)$, and $u(1-u) = 1/4 - (u - 1/2)^2$. It follows, by comparing the normalizing integral with its central part, that $\mathbb{P}(|V_n| > t) \leq C e^{-ct^2}$ in the relevant range, with harmless changes of constants near the endpoints. Conditional binomial exponential bounds give the same form for $Z_{i,n}$. In particular, all fixed moments of the Y_i are uniformly bounded, and for any fixed $0 < \eta < 1/2$,

$$\mathbb{P}\{\max_i |Y_i| > n^\eta\} \leq C_r e^{-c_r n^{2\eta}}. \quad (19.11)$$

These facts also follow from the exact moment formula in the next section.

[write] Added 5 October 2026, batch 98: explicit constants in (19.11). The phrase “with harmless changes of constants near the endpoints” can be replaced by explicit constants. With $\delta = u - \frac{1}{2}$, $1 - u \leq 1$ and the standard bound $\binom{2n}{n} \leq 4^n / \sqrt{\pi n}$,

$$f(u) \leq n \binom{2n}{n} \left(\frac{1}{4} - \delta^2\right)^{n-1} \leq 4\sqrt{n/\pi} e^{-4(n-1)\delta^2}.$$

Integrating over $|\delta| > s$ and using $(s+v)^2 \geq s^2 + v^2$ gives $\mathbb{P}(|U - \frac{1}{2}| > s) \leq 2\sqrt{n/(n-1)} e^{-4(n-1)s^2}$; with $s = t/(2\sqrt{n})$ and $n \geq 2$, $\mathbb{P}(|V_n| > t) \leq 2\sqrt{2} e^{-t^2/2}$ for all $t \geq 0$. Given $U = u$, Hoeffding’s inequality for the n Bernoulli variables of row i gives $\mathbb{P}(|Z_{i,n}| > t \mid U = u) \leq 2e^{-t^2/2}$, uniformly in u . Since $Y_i = Z_{i,n} + V_n$, it follows that $\mathbb{P}(|Y_i| > t) \leq 5e^{-t^2/8}$ for all $n \geq 2$ and $t \geq 0$. This gives the uniform moment bounds, and (19.11) with $C_r = 5r$ and $c_r = 1/8$, with no exception near the endpoints.

On the central event in (19.11), $\Psi_r \rightarrow 1$ with denominators uniformly separated from zero. Outside it, $W_n \leq 1$ makes its contribution to (19.4) smaller than any power of n^{-1} , even after multiplication by n^{d_R} . Therefore

$$\frac{H_m(n)}{M_m(n)} \sim \frac{J_r}{\binom{2n}{n} (4n)^{d_R} 9^r}. \quad (19.12)$$

Using Stirling's formula for $M_m(n)$ and $\binom{2n}{n}$ gives (14.6) and the leading term of Theorem 14.2. Finally (18.6) transfers it to $R_m(n)$. The power is

$$\frac{m-1}{2} - \frac{1}{2} + d_R = \frac{r}{2} + \binom{r}{2} = \frac{r^2}{2},$$

which explains the drop from the total-count exponent.

20 All-order coefficients and the universal first correction

20.1 Exact factorial moments of the mixture

For nonnegative integers $k_1, \dots, k_r$, let $K = k_1 + \dots + k_r$. Conditional binomial factorial moments and the beta integral give

$$\mathbb{E} \prod_{i=1}^r X_i^{k_i} = \left(\prod_{i=1}^r n^{k_i} \right) \frac{(n)^{\bar{K}}}{(2n+1)^{\bar{K}}}. \quad (20.1)$$

Here falling and rising factorials are distinguished typographically. Every joint ordinary moment of X is a finite rational function of n , by the Stirling-number identity $x^a = \sum_k \left\{ \begin{smallmatrix} a \\ k \end{smallmatrix} \right\} x^{\underline{k}}$. Therefore every joint moment of Y has an effective Laurent expansion in $\varepsilon = n^{-1/2}$. For a monomial of total degree A , only powers of ε of parity A occur. The bounded-moment argument above ensures that no negative powers survive.

Let $\mathcal{E}_\varepsilon$ be the linear functional on polynomials in $y_1, \dots, y_r$ obtained by replacing each monomial by this formal moment expansion. Set

$$Q_r(z) = \frac{1}{r! J_r} \mathcal{E}_\varepsilon \left[\Delta(y)^2 \Psi_r(\varepsilon, y) \right] \Big|_{\varepsilon^2=z}, \quad Q_r(0) = 1. \quad (20.2)$$

This is interpreted coefficient by coefficient, not as convergence of a formal infinite series. The symmetry $\Psi_r(-\varepsilon, -y) = \Psi_r(\varepsilon, y)$ shows that its coefficient of ε^k has polynomial parity k . Since Δ^2 has even total degree, the moment parity rule shows that the right-hand side of (20.2) contains only even powers of ε . Thus Q_r is a well-defined series in z with rational coefficients.

The remaining Stirling factor is

$$A_m(z) = \exp \left\{ \sum_{k \geq 1} \frac{B_{2k}}{2k(2k-1)} \left(m^{1-2k} - m + 2 - 2^{1-2k} \right) z^{2k-1} \right\}, \quad (20.3)$$

where B_{2k} are Bernoulli numbers. We obtain the explicit all-order rule

$$\boxed{b_{m,j}^R = [z^j] A_m(z) Q_{m-2}(z)}. \quad (20.4)$$

For a prescribed j , all operations are finite: expand Ψ through ε^{2j} , take finitely many moments from (20.1), and truncate (20.3) at z^j .

20.2 Why the formal calculation is an asymptotic theorem

Completion of the all-order part of Theorem 14.2. Fix L . On $\max |Y_i| \leq n^\eta$, choose any $0 < \eta < 1/2$. Each rational factor of (19.8) has denominator bounded away from zero for large n . Taylor expansion through order $2L + 1$ has a remainder bounded by

$$C_{r,L} \varepsilon^{2L+2} (1 + \|Y\|)^{N_{r,L}}$$

there, for some fixed exponent $N_{r,L}$. This follows either by finite geometric-series remainders in the displayed denominators, or by Taylor's formula on the line segment from 0 to ε ; throughout that segment $|\varepsilon Y_i|$ tends uniformly to zero. After multiplication by Δ^2 , uniform bounded moments bound the expected remainder by $O_{r,L}(n^{-L-1})$.

The complement of the central event contributes exponentially little to the original weighted expectation, by $0 \leq W_n \leq 1$ and (19.11). It also contributes exponentially little to each polynomial Taylor term, by the same tail bound and higher bounded moments. We may consequently replace each truncated polynomial expectation by its full exact expectation. Formula (20.1) expands these rational functions to the desired order with an ordinary rational remainder estimate. The parity argument removes half-integral orders. This proves the Poincaré expansion specified by $\mathcal{Q}_r$.

Stirling's formula with its fixed-order remainder yields A_m and (20.4). Finally

$$\frac{M_m(n)/\binom{3n}{n}}{M_m(n)/\{\binom{2n}{n} n^{d_R}\}} = n^{d_R} \frac{\binom{2n}{n}}{\binom{3n}{n}} = O_r\left(n^{d_R} (16/27)^n\right).$$

Thus the difference between R_m and H_m affects no coefficient of the rare-scale inverse-power expansion. All coefficients are rational by construction, and the case $r = 0$ is the exact identity already noted. $\square$

The theorem asserts effective coefficients and existence of fixed-order error constants. The package does not supply optimized numerical values for these constants or a rigorously certified onset of the asymptotic regime.

20.3 An exact Vandermonde identity

It is useful to derive the first correction without a large polynomial expansion. For $X \sim \text{Bin}(n, p)$ consider

$$F_X(t) = (1 + (1 - p)t)^x (1 - pt)^{n-x}. \quad (20.5)$$

The polynomial $P_j(x) = j! [t^j] F_X(t)$ is monic of degree j . Direct binomial expectation gives

$$\mathbb{E} F_X(t) F_X(s) = (1 + p(1 - p)ts)^n. \quad (20.6)$$

Hence the P_j are orthogonal and have squared norms $j! n^j [p(1 - p)]^j$. Replacing Vandermonde columns by these monic polynomials and expanding the determinant gives, for independent binomial variables conditional on $U = p$,

$$\mathbb{E}[\Delta(Y)^2 \mid U = p] = r! J_r \left(\prod_{j=0}^{r-1} \frac{n^j}{n^j} \right) [4p(1 - p)]^{d_R}. \quad (20.7)$$

Integrating the beta law,

$$\mathbb{E}[U^{d_R} (1 - U)^{d_R}] = \frac{(n)^{\overline{d_R}} (n + 1)^{\overline{d_R}}}{(2n + 1)^{2\overline{d_R}}}. \quad (20.8)$$

It follows by expanding these finite products that

$$\frac{\mathbb{E}\Delta(Y)^2}{r!J_r} = 1 - \frac{r(r-1)(2r-1)}{12n} + O_r(n^{-2}). \quad (20.9)$$

This identity isolates the correction caused by finite binomial and beta moments before inserting the rational interaction factors.

20.4 Evaluation of the interaction correction

Write $S_1 = \sum_i y_i$ and $S_2 = \sum_i y_i^2$. Expanding (19.8) begins with

$$\Psi_r(\varepsilon, y) = 1 + \varepsilon^2 P_1(y) + O(\varepsilon^3), \quad P_1(y) = \frac{9r-50}{36} S_2 + \frac{1}{4} S_1^2 - \frac{4r}{3}. \quad (20.10)$$

The $O(\varepsilon^3)$ notation here denotes a formal polynomial expansion; the previous proof supplies its expectation-level meaning.

Under the Gaussian law tilted by $\Delta(Z)^2$, two elementary identities are

$$\mathbb{E}_* \sum_i Z_i^2 = r^2, \quad \mathbb{E}_* \left(\sum_i Z_i \right)^2 = r. \quad (20.11)$$

For the first, scale the Gaussian integral: the degree of Δ^2 is $r(r-1)$ and the volume dimension is r , so differentiation in the Gaussian variance gives r^2 . For the second, separate the center of mass from its orthogonal complement. The Vandermonde is independent of the center of mass, whose sum coordinate is $N(0, r)$. Since $Y_i = Z_i + V$ and $\text{Var } V = 1/2$, with the tilt still independent of V ,

$$\mathbb{E}_* S_2 = r^2 + r/2, \quad \mathbb{E}_* S_1^2 = r + r^2/2. \quad (20.12)$$

Substitution into (20.10) yields

$$\mathbb{E}_* P_1 = \frac{r(9r^2 - 41r - 64)}{36}. \quad (20.13)$$

Combining with (20.9), the first coefficient of Q_r is

$$[z]Q_r(z) = \frac{r(3r^2 - 32r - 67)}{36}. \quad (20.14)$$

The first coefficient of A_m is

$$[z]A_m(z) = \frac{1}{8} + \frac{1/m - m}{12}. \quad (20.15)$$

Adding (20.14) and (20.15), with $m = r + 2$, gives (14.7). This proves the universal first correction independently of the higher-order implementation.

21 Height five, conditional shapes, and exponential separation

21.1 Three corrections for the A181199 rare subcount

Applying (20.4) at $m = 5$ gives

Corollary 21.1 (Height-five rare expansion). *For $R_5(n) = \#\{A : a_{1,n} < a_{5,1}\}$,*

$$R_5(n) = \frac{\sqrt{5}}{93312\pi^{3/2}} \left(\frac{3125}{4}\right)^n n^{-9/2} \times \left(1 - \frac{1393}{120n} + \frac{656449}{28800n^2} + \frac{9833844689}{155520000n^3} + O(n^{-4})\right). \quad (21.1)$$

The rational coefficients are finite evaluations of the proven formula, not a fit to sequence values. Appendix D describes the implementation-independent arithmetic used to reproduce them. For comparison, the first three corrections at adjacent heights are

m	$b_{m,1}^R$	$b_{m,2}^R$	$b_{m,3}^R$
2	0	0	0
3	$-199/72$	$-6479/10368$	$-491815/2239488$
4	$-979/144$	$74083/13824$	$25693405/1990656$
5	$-1393/120$	$656449/28800$	$9833844689/155520000$

Here R_5 is a newly considered refinement of the A181199 counting problem; it is not assigned a new OEIS identifier in this report.

21.2 How rare is the absence of a common active phase?

Let $Q_m(n) = T_{m,B^m D^m}(n)$ for $n \geq 2$. Then exactly

$$T_m(n) = Q_m(n) + R_m(n). \quad (21.2)$$

Both Q_m and R_m separately have rational-diagonal generating functions: each is a fixed finite sum of the event sectors in Theorem 14.1, up to the harmless terms at $n = 0, 1$. Using (14.2) from the preceding report gives

Corollary 21.2 (Rare-event probability). *For fixed $m \geq 2$, a uniformly chosen shifted rectangle satisfies*

$$\mathbb{P}(D_1 < B_m) = \frac{R_m(n)}{T_m(n)} \sim \frac{\sqrt{\pi} 4^{2m-3}}{9^{m-2}(m-2)!(m-1)!} n^{2m-5/2} 4^{-n}. \quad (21.3)$$

At height five, the first correction is

$$\frac{R_5(n)}{T_5(n)} = \frac{1024\sqrt{\pi}}{6561} n^{15/2} 4^{-n} \left(1 - \frac{509}{24n} + O(n^{-2})\right). \quad (21.4)$$

Proof. Divide the leading constants and powers in (14.5) and (14.2). The factorial quotient is $J_{m-2}/J_m = 1/((m-2)!(m-1)!)$, and the powers of 4 give 4^{2m-3} . For the correction at height five, the preceding report gives $b_{5,1}^{\text{total}} = 48/5$, so subtraction gives $-1393/120 - 48/5 = -509/24$. $\square$

Thus Q_m has the same dominant-scale inverse-power expansion as T_m to every fixed order, yet their difference has the explicitly identified exponential scale $(m^m/4)^n$. This is a combinatorially defined, beyond-all-orders refinement. It does *not* by itself identify a Stokes multiplier of a chosen analytic interpolation or summation of the dominant formal series. There may be other exponentially small structures inside Q_m , and the next boundary sector remains to be analyzed.

21.3 The middle shape conditioned on the rare event

The same argument gives a probabilistic limit theorem. Use the auxiliary uniform-sample model of Section 19. Condition on a legal array and on $D_1 < B_m$. Let X_i count middle-row entries below the top-row maximum U , and use the scaling (19.5).

Theorem 21.3 (Rare conditional shape). *For fixed $m \geq 3$, conditional on the rare event, the probability that all middle rows are active at the first completion tends to one. On that core event, the limiting joint law is*

$$(Y_1, \dots, Y_r, V_n) \Rightarrow (\Lambda_1 + V, \dots, \Lambda_r + V, V), \quad (21.5)$$

where $V \sim N(0, 1/2)$ is independent of $\Lambda_1 > \dots > \Lambda_r$, whose density on the decreasing chamber is

$$\frac{1}{J_r} \Delta(\lambda)^2 \prod_{i=1}^r \frac{e^{-\lambda_i^2/2}}{\sqrt{2\pi}}. \quad (21.6)$$

All fixed polynomial moments converge as well.

Proof. The first statement is (18.6) divided by the asymptotic for R_m . In the mixture representation, conditioning on legality and the core multiplies the joint law by $W_n(X)$ and restricts the counts to the decreasing chamber. After normalization, (19.7), (19.6), and the tail estimates show that the limiting weight is $\Delta(Z + V)^2 = \Delta(Z)^2$. It is independent of V , so the common Gaussian mode is not tilted. Restricting the Z coordinates to decreasing order changes the normalization from $r!J_r$ to J_r . The tail estimates with additional polynomial factors give moment convergence. The omitted noncore configurations have exponentially small conditional probability; their scaled coordinates are bounded by $O(\sqrt{n})$, so they do not affect any fixed moment. $\square$

The auxiliary U is a uniform-order-statistic realization of the combinatorial cut. The theorem is not a claim that a tableau itself contains real-valued sample times. In particular, its statement about Y_i , the scaled integer row lengths, is intrinsic; the joint statement with U uses the specified coupling.

22 Inverting the rare-count asymptotic profile

For $m \geq 3$ put

$$\lambda_m = m \log m - \log 4 > 0, \quad \beta_m = \frac{(m-2)^2}{2} > 0, \quad a = \log(y/\mathcal{K}_m). \quad (22.1)$$

The leading continuous profile is $y = \mathcal{K}_m e^{\lambda_m x} x^{-\beta_m}$. Its large solution is

$$x_0 = -\frac{\beta_m}{\lambda_m} W_{-1} \left(-\frac{\lambda_m}{\beta_m} e^{-a/\beta_m} \right), \quad (22.2)$$

where W_{-1} is the real lower branch. The argument belongs to $(-1/e, 0)$ for sufficiently large y . The branch is forced by the requirement $x_0 \rightarrow \infty$; the principal branch gives the small solution instead. Lambert-core inversion and inverse-power reversion are standard tools, also used for the total sequence in Part I (Section 10). Their application here uses the different rare-event rate and power.

Proposition 22.1 (Two corrections to the inverse profile). Write $b_1 = b_{m,1}^R$, $b_2 = b_{m,2}^R$, $\lambda = \lambda_m$, and $\beta = \beta_m$. The formal large inverse has

$$x = x_0 - \frac{b_1}{\lambda x_0} - \frac{1}{x_0^2} \left(\frac{b_2 - b_1^2/2}{\lambda} + \frac{\beta b_1}{\lambda^2} \right) + O(x_0^{-3}). \quad (22.3)$$

Every subsequent coefficient is obtained recursively from (20.4). For any fixed sufficiently long truncation of the real asymptotic profile, its genuine large real inverse has the corresponding finite expansion.

Proof. Take logarithms:

$$\lambda x - \beta \log x + \log \left(1 + b_1/x + b_2/x^2 + \cdots \right) = a.$$

Substitute $x = x_0 + c_1/x_0 + c_2/x_0^2 + \cdots$ and use $\lambda x_0 - \beta \log x_0 = a$. At orders x_0^{-1} and x_0^{-2} ,

$$\lambda c_1 + b_1 = 0, \quad \lambda c_2 - \beta c_1 + b_2 - b_1^2/2 = 0.$$

This gives (22.3). At every later order the next unknown coefficient occurs linearly with nonzero coefficient λ . For a fixed real profile truncation, its logarithmic derivative tends to $\lambda > 0$. Taylor's theorem or the real implicit function theorem then turns the recursive formal truncation into the stated inverse asymptotic. $\square$

For height five, $b_2 - b_1^2/2 = -535/12$. Thus, with $\lambda = \log(3125/4)$ and $\beta = 9/2$,

$$x = x_0 + \frac{1393}{120\lambda x_0} + \frac{1}{x_0^2} \left(\frac{535}{12\lambda} + \frac{4179}{80\lambda^2} \right) + O(x_0^{-3}). \quad (22.4)$$

The leading logarithmic scale can also be read without W :

$$x_0 = \frac{1}{\lambda} \left(a + \beta \log a - \beta \log \lambda + O\left(\frac{\log a}{a}\right) \right).$$

22.1 A discrete inverse is a staircase

The sequence $R_m(n)$ is eventually strictly increasing for $m \geq 3$, since its successive ratio tends to $e^{\lambda_m} > 1$. Define $N_m(y) = \min\{n \geq 1 : R_m(n) \geq y\}$ for large y . Let $\xi_L(y)$ be the large inverse of the truncated continuous profile

$$\mathcal{K}_m e^{\lambda_m x} x^{-\beta_m} \sum_{j=0}^L b_{m,j}^R x^{-j}.$$

The fixed-order remainder and a logarithmic derivative bounded away from zero imply that, for some $C_{m,L}$ and all sufficiently large y ,

$$\left[\xi_L(y) - C_{m,L} \xi_L(y)^{-L-1} \right] \leq N_m(y) \leq \left[\xi_L(y) + C_{m,L} \xi_L(y)^{-L-1} \right]. \quad (22.5)$$

Indeed, the asymptotic relative error is $O(n^{-L-1})$, so the two smooth comparison profiles cross the target within that horizontal distance. When the two displayed ceilings coincide, their common value is the threshold. No such agreement can be asserted uniformly near integer crossings without explicit error constants and distance information. The package's numerical inverse checks are therefore profile diagnostics, not integer-threshold certificates.

[write] Added 5 October 2026, batch 98: the inversion is an instance. As for Part I's (10.3) (note at the end of Section 10), (22.2) is a case of Theorem J.23 (p0:thm:lambert-core) of the transseries volume Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/, with $a = \lambda_m > 0$ and $b = -\beta_m < 0$, whose branch rule for $b < 0$ selects W_{-1} ; the caution about integer thresholds and (22.5) are the separation condition of that volume's staircase theorem J.43 (p0:thm:staircase). The manuscript itself calls these tools standard. Two steps are argued in one sentence each and are recorded in Section 24.1: the last sentence of Proposition 22.1 (that the genuine real inverse has the formal expansion) and the staircase (22.5), whose constant $C_{m,L}$ and range of validity are existential.

23 Exact checks, numerical diagnostics, and reproducibility

23.1 Independent computations

The package includes three genuinely different finite enumerations: unit-step prefix-state dynamic programming; an independent bitmask order-ideal enumerator built directly from the cell comparisons; and an event-skeleton evaluator using the signed multinomial kernel. The hook core sum and the symbolic beta-moment expansion are separate again. No candidate OEIS recurrence is used to generate a test value.

The verification script completed **1,954 exact checks**. The categories and counts are recorded in `data/verification.json`:

Check	Number
Independent poset enumeration versus prefix states	28
Reflection kernel versus independent chamber walks	887
Event decomposition versus prefix states	25
Event-word realizability and rotation symmetry	640
Rare event-word sum and the worked example	21
OEIS reference values (heights four and five)	20
Core lower bound, separation bound, and core error bound	228
Classical Catalan and small-width controls	64
Universal first correction and saved higher coefficients	7
Common-mode Gaussian moments and exact beta moments	34
Total	1954

The reference terms are small data excerpts from A181198 and A181199, with source URLs and access date in the audit. The tests regenerate these values rather than assuming their recurrence. Finite checks corroborate the arguments; they are not proofs of an infinite identity or a substitute for peer review.

23.2 Accuracy on the rare exponential scale

Let $\mathcal{L}_5(n)$ denote the leading term in (21.1), and put $\mathcal{A}_{5,L}(n) = \mathcal{L}_5(n) \sum_{j=0}^L b_{5,j}^R n^{-j}$. The following entries are relative errors $\mathcal{A}_{5,L}(n)/R_5(n) - 1$ computed against exact integer rare counts.

n	$L = 0$	$L = 1$	$L = 2$	$L = 3$
10	4.46198	-1.87847	-0.633499	-0.288127
20	1.05994	-0.135683	-0.0183006	-0.00201881
40	0.379221	-0.0210405	-0.00139232	-0.0000296524
60	0.229776	-0.00815163	-0.000365317	-0.00000531141

These values illustrate both the usefulness of the corrections and the slow onset of the leading asymptotic at height five. The negative first-order approximation at $n = 10$ is not a contradiction: fixed-order asymptotic expansions are not guaranteed positive or accurate at small n .

The core already captures the rare count particularly well at moderate width:

n	$R_5(n)/\mathcal{L}_5(n)$	$H_5(n)/R_5(n)$
10	0.183083833013	0.862684900894
20	0.485450790529	0.999319561936
40	0.725047021487	0.999999971834
60	0.813156328776	0.99999999998899

The underlying integer values and 70-digit calculations are provided in `data/numerics.csv`. Displayed decimals are diagnostics, not interval enclosures.

23.3 Reproducing the package

From the package root, a Python environment with SymPy and mpmath can run

```
python code/verify.py
python code/coefficients.py --height 5 --order 3
python code/numerics.py
python code/inverse_profile.py --index 60 --height 5
```

The first command reruns the exact tests and writes their log. The second regenerates the three height-five rational corrections. The third regenerates the decimal comparison tables from exact rare counts. The fourth checks the inverse profile at a target obtained from an exact rare count; it is not a threshold-certification procedure. The tested environment is Python 3.13.5, SymPy 1.14.0, and mpmath 1.3.0. The standard-library enumerators themselves do not require either symbolic or numerical package. No Maple diagonal-elimination computation, Lean proof, or Rocq proof is claimed.

[write] Added 5 October 2026, batch 98: the shipped layout. “The package” in this section is the delivered archive. In Provelt its files carry the prefix `02-boundary-phases-`:

Delivered as	Shipped as
<code>code/name.py</code> (five scripts)	<code>code/02-boundary-phases-name.py</code>
<code>build.sh</code>	<code>code/02-boundary-phases-build.sh</code>
<code>data/name</code> (nine records)	<code>data/02-boundary-phases-name</code>
<code>requirements.txt</code>	<code>data/02-boundary-phases-requirements.txt</code>
<code>notes/PROPOSED_OEIS_NOTE.md</code>	<code>data/02-boundary-phases-notes-PROPOSED_OEIS_NOTE.md</code>
<code>notes/SOURCES_AND_STATUS.md</code>	<code>02-boundary-phases-notes-SOURCES_AND_STATUS.md</code>

The scripts import one another and read and write `data/` under their delivered names (`verify.py` imports `tableaux`, and every script sets its root to the parent of `code/`), and `build.sh` changes to its own directory; so they run only on a copy with the delivered names restored, never in place, and the README of the report gives the commands. At intake (5 October 2026, on such a copy; Python 3.13.5, SymPy 1.14.0, mpmath 1.3.0) `verify.py` passed all 1,954 checks in 66 s (the package recorded 23 s), `numerics.py`, `coefficients.py -height 5 -order 3` and `inverse_profile.py -index 60 -height 5` reproduced the shipped records after removal of carriage returns, and the only differences were the timing fields of `verification.json`, `verification.txt` and `numerics.json`. The check table above merges some of the script’s sixteen categories (its $640 = 320 + 320$, $21 = 20 + 1$, $228 = 3 \cdot 76$, $7 = 4 + 3$ and $34 = 6 + 28$); the total agrees.

`code/02-boundary-phases-verify.py` embeds the first ten terms of A181198 and A181199 as reference values. They are OEIS content, published by The OEIS Foundation Inc. under the Creative Commons Attribution-ShareAlike 4.0 licence, so those values are third-party data under that licence, not MIT-0. The draft `data/02-boundary-phases-notes-PROPOSED_OEIS_NOTE.md` is headed “NOT SUBMITTED” and stays so: nothing was submitted to OEIS by the package’s author or by the intake, and its instruction not to remove the conjectural status of the displayed recurrences stands.

24 Further research questions and concrete next steps

Research question 8 (Certify the particular low-order recurrences). Convert the event sum at heights four and five to a reduced rational integral, compute a telescoper, and compare it with the conjectured OEIS operators. A successful certificate must also control singular initial indices. This would settle a stronger statement than the existence theorem proved here and could resolve Conjectures 18 and 19 of [7].

Research question 9 (Reduce the dimension of the exact formula). The bound $m(m-1)$ is deliberately structural rather than optimal. Can adjacent chamber kernels be convolved before summation, or can a Pfaffian/determinantal identity absorb the ordered births and deaths? A substantially smaller representation would make the constructive D-finiteness theorem computationally practical at larger heights.

Research question 10 (Determine deeper boundary sectors). The error in (18.6) contains arrays for which two or more bottom rows are unborn at the first completion. Determine its sharp rate, constant, and all-order expansion, then stratify by the number of unborn rows and early completions. The crude multinomial bound supplies an upper rate, not the next asymptotic term.

Research question 11 (Find minimal differential complexity). How do the minimal recurrence order, polynomial degree, and minimal rational-diagonal dimension grow with m ? Does the event decomposition impose a useful uniform bound much smaller than a general elimination bound? The conjectured small orders at heights four and five suggest substantial cancellation that the existence proof does not explain.

Research question 12 (Describe event-rank limit laws). Corollary 17.2 gives exact D-finite moment totals. Use these or the beta representation to derive joint laws for B_i, D_i , both unconditionally and under $D_1 < B_m$. Determine whether the one-time conditional shape theorem extends to a process of interacting paths near the first completion.

Research question 13 (Make height growth uniform). The proofs fix m . Determine ranges $m = m(n)$ in which the Gaussian reduction and rare-sector asymptotics remain uniform. The dimensions of the Vandermonde and the interaction product grow quadratically, so fixed height error estimates cannot simply be reused. Identify the crossover between a probable common active phase and a boundary-dominated regime.

Research question 14 (Obtain certified finite-width bounds and inversion). Replace the existential remainders in (14.5) by computable, useful bounds with explicit onset. Combine them with (22.5) to certify integer thresholds, including inputs very close to an integer crossing. This requires more than high-precision floating-point evaluation of the asymptotic series.

Research question 15 (Generalize and formalize the finite-event method). Investigate unequal row lengths and other fixed-offset antidiagonal constraints whose active states may still be described by a finite collection of reflection chambers. Separately, formalize the prefix-state bijection, reflection involution, and binomial-sum encoding in a proof assistant. Each is a finite combinatorial component with a clear interface; a formal proof of the analytic rare-event expansion is a larger second stage.

[write] Added 5 October 2026, batch 98: the questions of Part II and Part I. The eight questions above are the manuscript’s Research questions 1–8, numbered 8–15 here. Question 8 refines Part I’s Research question 1 (a telescoper route from the event sum, and Conjectures 18–19 of [7]); Question 11 is the open second half of Part I’s Research question 7; Questions 12, 13 and 14 continue Research questions 5, 3 and 6 of Part I.

24.1 Further questions and research recorded at the write

Added 5 October 2026, batch 98, under Vladimir Reshetnikov’s standing rule that unproved claims are recorded as open questions rather than dropped. Each item names its source, the argument it came with, and what is missing.

- (F1) *Other exponentially small structures* (the manuscript, after Corollary 21.2). The text says there “may be other exponentially small structures inside Q_m , and the next boundary sector remains to be analyzed”. No argument is offered either way; Proposition 18.3 shows that the boundary error of Part I’s central cut has rate 2^{-n} , far above the rare rate, but that error is an artefact of the cut, not a sector of T_m . Missing: a classification of the event sectors of Theorem 16.1 by exponential rate.
- (F2) *All orders for the boundary error* (the write; Research question 4 of Part I, re-scoped). Proposition 18.3 gives the leading term of $E_{m,n}(1/2)$. Sketch: by (18.7) the leading stratum is an interior sum with $m - 1$ independent binomial rows and the extra factor $\prod_i (1 + \varepsilon Y_i)/(3 - \varepsilon Y_i)$, whose odd powers of ε cancel in expectation by the symmetry $Y \mapsto -Y$; Section 5’s moment polynomials and Taylor control should give an expansion in powers of $1/n$. Missing: writing that argument, computing its coefficients, and the analogous analysis of the $O(4^{-n})$ strata $D_{m,n}$ and of how the interior sum and the boundary terms combine.
- (F3) *Sketch-level steps of Sections 19, 21 and 22*. The joint convergence (19.6) and the tail bound (19.11) were sketches; the write’s notes after them supply complete arguments. Still argued only in outline: the moment convergence in Theorem 21.3 (“the tail estimates with additional polynomial factors give moment convergence”; with the explicit tails it reduces to uniform integrability of $|Y|^k W_n(X)/\mathbb{E}W_n(X)$, which needs $W_n \leq Cn^{-d_R} \Delta(Y)^2$ on the central event and the exponential tails off it); the last sentence of Proposition 22.1 (the genuine real inverse has the formal expansion: a real implicit-function argument, as in Theorem J.23 of the transseries volume); and the staircase (22.5), whose constant $C_{m,L}$ and threshold for “sufficiently large y ” are existential (Research question 14 asks for explicit ones).
- (F4) *Priority of Theorem 14.1* (the manuscript and its source audit). The audit searched the OEIS entries, Kauers–Koutschan, Bostan–Lairez–Salvy, Sun and the web for the sequence identifiers with “proved”, “D-finite”, “shifted rectangles” and “diagonal”, and found no earlier proof; it did not search the general literature on D-finiteness of linear-extension counts of fixed-height posets or of walks in Weyl chambers, where the theorem might be a known special case. Missing: that search.

25 Conclusion

The obstruction in the shifted-rectangle model is not the strict interior walk but the coexistence of births, completions, and two kinds of boundary ties. Recording those events turns the exact enumeration into a finite binomial construction. That proves rational diagonality and D-finiteness for every fixed height without a guessed recurrence.

The same boundaries also carry information invisible to the dominant inverse-power expansion. The absence of a common active phase has rate 4^{-n} relative to the main sequence, an explicit polynomial multiplier, and its own all-order expansion. A random first-completion cut exposes both the factor 9^{-r} and a Gaussian common mode independent of the Vandermonde repulsion. These statements are precise additions to the

preceding asymptotic report. They leave clear next targets: compact operators, the specific OEIS recurrence certificates, and deeper exponential boundary strata.

A A direct proof of the shifted hook product

Let $\lambda_1 > \dots > \lambda_r > 0$ be a strict partition of k , and put

$$H(\lambda) = \frac{k!}{\prod_i \lambda_i!} \prod_{i < j} \frac{\lambda_i - \lambda_j}{\lambda_i + \lambda_j}.$$

We prove that it satisfies the strict-partition path recursion

$$H(\lambda) = \sum_i H(\lambda - e_i), \quad (\text{A.1})$$

where only valid strict-partition removals are included and a trailing zero part is deleted. Together with $H(\emptyset) = 1$, this proves (3.5) by induction.

For a valid removal,

$$\frac{H(\lambda - e_i)}{H(\lambda)} = \frac{\lambda_i}{k} \prod_{j \neq i} \frac{\lambda_i - \lambda_j - 1}{\lambda_i - \lambda_j} \frac{\lambda_i + \lambda_j}{\lambda_i + \lambda_j - 1}. \quad (\text{A.2})$$

The same expression is zero for an invalid removal causing equal positive parts. If the last part is one, deleting its resulting zero gives the same ratio, since the extra zero factors cancel. Thus it suffices to prove

$$\sum_i \lambda_i \prod_{j \neq i} \frac{\lambda_i - \lambda_j - 1}{\lambda_i - \lambda_j} \frac{\lambda_i + \lambda_j}{\lambda_i + \lambda_j - 1} = k. \quad (\text{A.3})$$

Consider the rational function

$$R(z) = \prod_{j=1}^r \frac{z - \lambda_j - 1}{z - \lambda_j} \frac{z + \lambda_j}{z + \lambda_j - 1}, \quad f(z) = (z - \tfrac{1}{2})R(z).$$

Its possible finite poles are the disjoint sets λ_i and $1 - \lambda_i$. Direct substitution gives $R(1 - z) = R(z)$ and hence $f(1 - z) = -f(z)$. The residues at the paired poles are equal. At $z = \lambda_i$, the residue is the negative of the i th summand in (A.3).

At infinity, each pair of factors contributes $-2\lambda_j/z^2$ to the first nonconstant term, so

$$R(z) = 1 - \frac{2k}{z^2} + O(z^{-3}), \quad f(z) = z - \frac{1}{2} - \frac{2k}{z} + O(z^{-2}).$$

The sum of finite residues is the coefficient of z^{-1} , namely $-2k$. On the other hand it is minus twice the left side of (A.3). The identity follows, proving (A.1) and the hook product. All operations here are rational identities; the residue language can equivalently be replaced by partial fractions.

B An implementation-independent coefficient specification

For a fixed height m and requested order L , the following finite recipe fully specifies the coefficient calculation.

First compute $h_0, \dots, h_L$ from $\prod_{i < j} (1 - u(y_i + y_j)^2)^{-1}$ modulo u^{L+1} . Next compute the sign cumulants κ_r from

$$\log \cosh t = \sum_{r \geq 1} \kappa_r \frac{t^r}{r!}.$$

The even values begin $\kappa_2 = 1$, $\kappa_4 = -2$, $\kappa_6 = 16$, $\kappa_8 = -272$; the odd values vanish. The moment polynomials can be computed recursively as

$$\mu_0(z) = 1, \quad \mu_k(z) = \sum_{\substack{2 \leq r \leq k \\ r \text{ even}}} \binom{k-1}{r-1} \kappa_r z^{r/2-1} \mu_{k-r}(z) \quad (k > 0), \quad (\text{B.1})$$

with $\mu_k = 0$ for odd k . Apply (11.1) to each h_r , evaluate each monomial by (5.8), retain powers through z^{L-r} , and combine them as in (5.9). Finally apply the formal exponential (5.16). This recipe is sufficient to reproduce the printed coefficients independently of the shipped Python code.

For $m = 5$, the two outputs through order five are

$$(c_{5,j})_{j=0}^5 = \left(1, 10, \frac{225}{4}, \frac{1819}{8}, \frac{22045}{32}, 1353\right),$$

$$(b_{5,j})_{j=0}^5 = \left(1, \frac{48}{5}, \frac{5233}{100}, \frac{1028391}{5000}, \frac{60248377}{100000}, \frac{273939939}{250000}\right).$$

The first two nonconstant entries in each row also have the independent symbolic derivation in Section 6. Higher printed entries are exact evaluations of the proved finite formula, not inferred from observed numerical errors.

C A fully finite recipe for the exact sum

[write] Added 5 October 2026, batch 98. Appendices C and D belong to Part II: they are the manuscript's Appendices A and B.

For clarity, the diagonal theorem can be implemented without making any choices suggested by numerical data. The following recipe has fixed size once m is fixed.

First list all Dyck words of semilength m . For a chosen word, make one vector v_s of length r_s for each interval $0 \leq s < 2m$, and replace the top coordinate before each completion by $n - 1$. Sum every remaining coordinate from 0 to n . Construct $u_0 = ()$ and the later u_s using (16.1). Insert H factors for

$$v_{s,r_s} \geq 1, \quad v_{s,1} \leq n - 1, \quad v_{s,i} - v_{s,i+1} \geq 1,$$

where the conditions with coordinates are omitted when $r_s = 0$. For a birth with $r_s > 0$, also insert $H(v_{s,r_s} - 2)$. The already substituted completion coordinate enforces its event condition.

For each interval independently choose a permutation $\sigma_s \in \mathfrak{S}_{r_s}$, multiply its sign, and set

$$a_{s,i} = v_{s,i} - u_{s,\sigma_s(i)}.$$

Multiply the binomial product (16.5) for this displacement vector. Sum over the finitely many permutations and endpoint coordinates, and then over the words. This expression equals $T_m(n)$ for every $n \geq 2$ by the bijective proof of Theorem 16.1.

The endpoint-coordinate order in this recipe can be arbitrary; using nested partial sums and then affine substitution of their upper bounds realizes the rectangular sums in the binomial-sum algebra. Values outside

the intended polytope are killed by the explicit H factors. In particular, one never applies a recurrence or divides by a factor that could vanish at a summation endpoint.

The delivered evaluator uses the same formula more efficiently by iterating over decreasing endpoint tuples directly and caching suffix computations. That optimization changes neither the mathematical formula nor its fixed-dimensional binomial nature. It is not claimed to beat the prefix dynamic program for every input.

D An implementation-independent coefficient algorithm

The symbolic routine can be reproduced using rational polynomial arithmetic alone. Here are sufficient details to fix every convention. For a nonnegative integer a , form the polynomial in p

$$C_a(n, p) = \sum_{b=0}^a \sum_{k=0}^b \binom{a}{b} 2^b (-n)^{a-b} \left\{ \begin{matrix} b \\ k \end{matrix} \right\} n^k p^k. \quad (\text{D.1})$$

It is exactly $\mathbb{E}[(2X - n)^a \mid U = p]$. For a monomial $y_1^{a_1} \cdots y_r^{a_r}$, set $A = \sum_i a_i$ and multiply $\prod_i C_{a_i}(n, p) = \sum_{k=0}^A c_k(n) p^k$. Then

$$\mathbb{E} \prod_i Y_i^{a_i} = n^{-A/2} \sum_{k=0}^A c_k(n) \frac{(n)^{\bar{k}}}{(2n+1)^{\bar{k}}}. \quad (\text{D.2})$$

A convenient common denominator is $(2n+1)^{\bar{A}}$; its numerator is

$$\sum_{k=0}^A c_k(n) (n)^{\bar{k}} (2n+1+k)^{\overline{A-k}}. \quad (\text{D.3})$$

Reverse the numerator and denominator polynomials and perform ordinary power-series division at $1/n = 0$, retaining the extra factor $n^{-A/2}$. This is the exact moment functional $\mathcal{E}_\varepsilon$ used in (20.2). It avoids repeated simplification of many separate rational functions.

To expand the single-coordinate interaction in (19.8), write

$$\frac{1 - (2 + y^2)\varepsilon^2 - 2y\varepsilon^3}{1 - (6 + y^2)\varepsilon^2/9 + 2y\varepsilon^3/9} = \sum_{k \geq 0} c_k(y) \varepsilon^k. \quad (\text{D.4})$$

With $c_k = 0$ for $k < 0$, its coefficients satisfy

$$c_k = N_k + \frac{6 + y^2}{9} c_{k-2} - \frac{2y}{9} c_{k-3}, \quad N_0 = 1, \quad N_2 = -(2 + y^2), \quad N_3 = -2y, \quad (\text{D.5})$$

and all other N_k are zero. The pair interaction contributes the geometric series

$$\sum_{k \geq 0} \frac{(y_i + y_\ell)^{2k}}{4^k} \varepsilon^{2k}, \quad i < \ell.$$

Multiply these finitely truncated factors, then multiply by $\Delta(y)^2$. Apply (D.2) to each monomial, divide by $r!J_r$, and finally multiply by the truncated exponential (20.3). No asymptotic fitting or guessed differential equation enters this calculation. For the height-five coefficients in (21.1), a truncation at ε^6 suffices.

As intermediate checks, the engine gives

$$[z]Q_1 = -\frac{8}{3}, \quad [z]Q_2 = -\frac{119}{18}, \quad [z]Q_3 = -\frac{34}{3},$$

which also follow from (20.14). The package stores exact coefficient strings and reruns their derivation in `code/verify.py`.

References

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